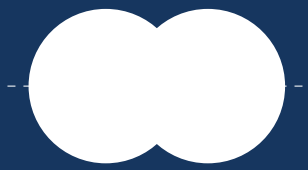
JEE Main & JEE Advanced | Inorganic / Physical Chemistry

CHEMICAL BONDING



σ overlap

Formula • Derivation • Conditions • Master Atlas

Complete derivation companion covering the ionic, covalent and co-ordinate bond, Valence Bond Theory, hybridisation, bond parameters, dipole moments and Molecular Orbital Theory

Contents of this atlas

Chemical bond and the cause of chemical combination (energy criterion and octet rule) · classification of bonds · ionic bond, Born–Haber cycle, lattice energy, solubility · Fajans' rules and polarisation · covalent and co-ordinate bonds, variable valency · Lewis structures and formal charge · VSEPR theory · Valence Bond Theory and orbital overlap · hybridisation (sp to sp^3d^3), Bent's rule, Drago generalisation · bond length, bond angle, bond energy · odd-electron species and non-existent molecules · dipole moment and percentage ionic character · Molecular Orbital Theory of homonuclear diatomics.

Reference source: ALLEN JEE Inorganic Chemistry (Nurture Course) – *Chemical Bonding*, theory booklet.
Design reference: *Periodic Table & Periodicity – Formula, Derivation, Conditions, Master Atlas.*
Method: every boxed result is preceded by its derivation or justification and followed by the conditions under which it applies; all figures, graphs and tables in this book are generated in code.

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Conventions used throughout

Bond enthalpies and lattice enthalpies are quoted in $\mathrm{kJ\,mol^{-1}}$ ($1\ \mathrm{eV} = 96.485\ \mathrm{kJ\,mol^{-1}}$); lengths in picometres ($1\ \mathrm{pm} = 10^{-12}\ \mathrm{m} = 0.01\ \text{Å}$) and dipole moments in debye ($1\ \mathrm{D} = 3.336 \times 10^{-30}\ \mathrm{C\,m}$). The *lattice energy* U and the *electron gain enthalpy* (EGE) are used, as in the ALLEN notes, as positive quantities of energy *released*; the sign of every term is therefore written explicitly in the Born–Haber relation. χ is Pauling electronegativity; Z_{eff} is effective nuclear charge; V, L, B in formal-charge work are valence electrons, lone-pair (non-bonding) electrons and bonding electrons respectively. Every boxed result is a final exam-ready formula; each is preceded by its derivation or justification and followed by the conditions under which it applies.

Source notes and corrections

Where the ALLEN notes contain a typographical slip or an out-of-date value, the corrected value is used here and flagged where it occurs: the bond length of F_2 is 144 pm (not 42 pm), the bond dissociation enthalpy of H_2 is $\approx 436\ \mathrm{kJ\,mol^{-1}}$ (the notes quote both 434.2 and 438), and the mass-spectrometric lattice values of Part II are standard Born–Haber data.

Part I — Foundations

1 Introduction

Except for the noble gases, no element exists as independent atoms under ordinary conditions; most exist as molecules, which are clusters of atoms. This chapter answers two questions: *why* do atoms combine, and *how* are the resulting bonds described?

1.1 Chemical Bond

- A chemical bond is a force that acts between two or more atoms and holds them together as a stable molecule.
- It is a union of atoms involving a *redistribution of electrons* among them.
- The process is accompanied by a *decrease in energy*; the magnitude of this decrease is a measure of the strength of the bond.
- A molecule forms only if it is more stable, i.e. lower in energy, than the separated atoms.

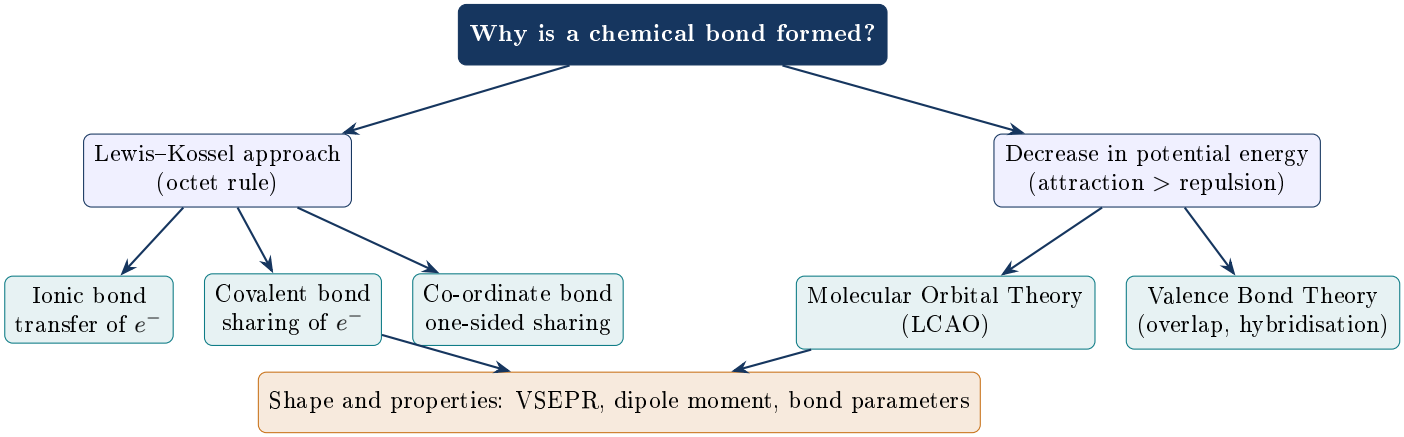


Figure 1. Flowchart to chemical bonding: the two reasons for bond formation (octet completion and energy lowering) lead to the classical bond types and to the two quantum-mechanical theories.

1.2 Cause of Chemical Combination

Tendency to acquire minimum energy

When two atoms approach each other, (a) the nucleus of each attracts the electrons of the other, while (b) the two nuclei repel each other and the electrons repel each other. If the net result is attraction, the energy of the system falls and a bond forms. Hence

$$\text{attraction} \propto \frac{1}{\text{energy}} \propto \text{stability},$$

and bond formation is always an *exothermic* process.

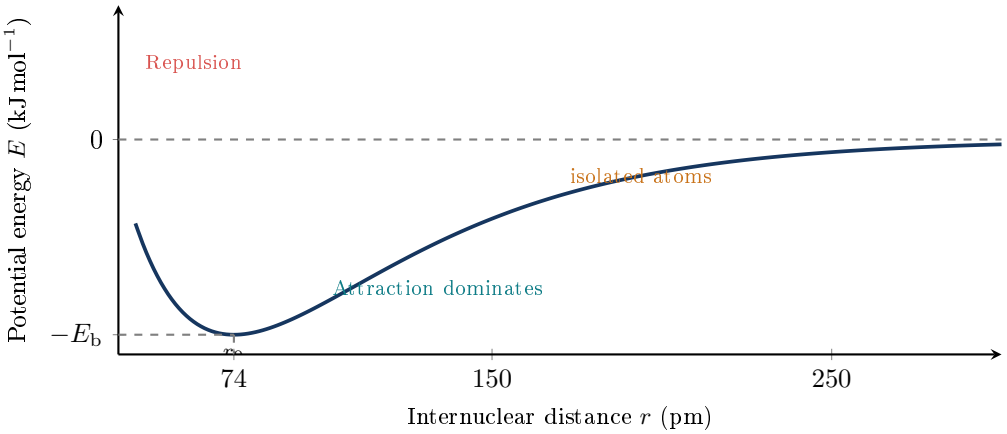


Figure 2. Potential-energy diagram for bond formation (Morse form, H_2: $r_0 = 74$ pm, well depth ≈ 436 kJ mol^{-1}). The system reaches its minimum energy where attraction and repulsion balance.

Octet rule

Proposed by Lewis and Kossel: atoms combine so as to complete an octet (s^2p^6) in the outermost shell, because a complete shell is the most stable state. The octet can be completed in three ways.

Mode	Description	Examples
Transfer of electrons	Complete transfer from one atom to another: **ionic bond**	NaCl, $CaCl_2$, MgO
Mutual sharing	Equal contribution of electrons by both atoms: **covalent bond**	Cl_2, N_2, O_2
One-sided sharing	Shared pair supplied by only one atom: **co-ordinate bond**	NH_4^+, $NH_3 \rightarrow BF_3$

Exceptions to the octet rule

Class	Explanation	Examples
(i) Transition-metal ions	Partly filled d sub-shell: [Ar]$3d^n$, no s^2p^6 shell	Cr^{3+} $3d^3$ (2,8,11); Mn^{2+} $3d^5$ (2,8,13); Fe^{2+} $3d^6$ (2,8,14)
(ii) Pseudo-inert gas configuration	Outer shell $s^2p^6d^{10}$ (18 electrons)	Zn^{2+} $3d^{10}$; Cd^{2+} $4d^{10}$
(iii) Contraction (incomplete) octet	Central atom has fewer than 8 electrons; electron-deficient	BeF_2 (4e), BF_3, $AlCl_3$, BCl_3 (6e)
(iv) Expansion of octet	Central atom of period ≥ 3 uses vacant d-orbitals	PCl_5 (10e), SF_6 (12e), ClF_3 (10e), ICl_5 (12e), IF_7 (14e)
(v) Odd-electron species	Odd total number of valence electrons	NO, NO_2, ClO_2
(vi) Compounds of noble gases	Complete octet (or duplet) yet compounds exist	XeF_2, XeF_6, KrF_2

Common Trap

Second-period elements (B, C, N, O, F) can never exceed an octet because they have no *d*-orbitals in the valence shell. Hence NF_5, OF_4, OF_6 do not exist, whereas PF_5, SF_4, SF_6 do.

2 Classification of Bonds

Attractive forces between atoms and molecules are divided into **strong bonds** (ionic, covalent, co-ordinate, metallic) and **weak interactions** (hydrogen bond, van der Waals interaction).

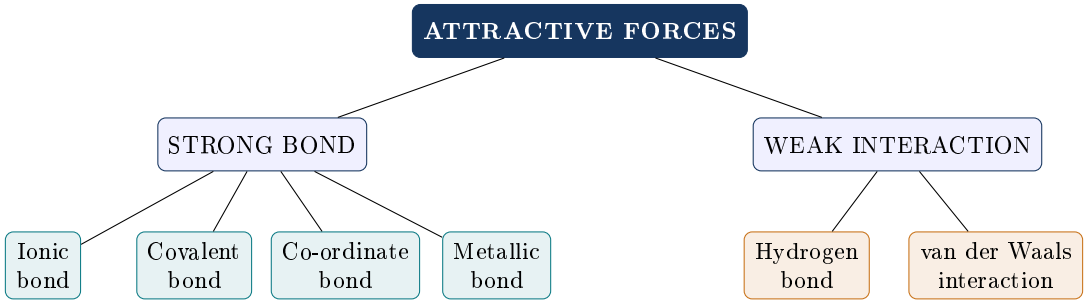


Figure 3. Classification of attractive forces.

Interaction	Origin	Typical energy
Ionic	Electrostatic attraction between cation and anion	400–4000 kJ mol^{-1} (lattice)
Covalent / co-ordinate	Sharing of an electron pair (overlap of orbitals)	150–950 kJ mol^{-1}
Metallic	Positive kernels in a sea of delocalised electrons	100–800 kJ mol^{-1}
Hydrogen bond	A–H$^{\delta+}\cdots$:B$^{\delta-}$ with A, B = F, O, N	$<$ 25 (weak), 25–35 (medium), $>$ 35 kJ mol^{-1} (strong; HF_2^- $\approx$ 220 kJ mol^{-1})
van der Waals	Dipole and induced-dipole attractions	0.1–20 kJ mol^{-1}

Hydrogen bond. When hydrogen is bonded to a strongly electronegative atom X (F, O, N), the shared pair is displaced towards X, so H acquires a partial positive charge and attracts the lone pair of a neighbouring electronegative atom. Intermolecular hydrogen bonding causes association (HF and H_2O are liquids while HCl and H_2S are gases); intramolecular hydrogen bonding (e.g. *o*-nitrophenol) lowers the boiling point and water solubility relative to the *m*- and *p*-isomers, which associate intermolecularly.

Part II — The Ionic Bond

3 Electrovalent or Ionic Bond

An ionic (electrovalent) bond is the electrostatic force of attraction between a cation and an anion formed by the *complete transfer* of one or more electrons. The electropositive atom (groups 1–13) loses electrons and the electronegative atom (groups 15–17) gains them. The total number of electrons lost or gained is the **electrovalency**.

Compound	Change	Electrovalency of metal	of non-metal
MgO	$Mg + O \longrightarrow Mg^{2+} + O^{2-}$ (2,8,2 and 2,6)	2	2
$CaCl_2$	$Ca + 2\,Cl \longrightarrow Ca^{2+} + 2\,Cl^-$	2	1
CaO	$Ca + O \longrightarrow Ca^{2+} + O^{2-}$	2	2

Features. The force acts equally in all directions, so the ionic bond is *non-directional*; a definite three-dimensional crystal lattice results; ionic compounds therefore have no molecular formula, only an empirical

(formula-unit) formula, e.g. NaCl.

3.1 Variable Electrovalency

s-block metals show a single electrovalency, while d- and p-block metals show variable electrovalency, for two reasons.

Inert pair effect

Down a p-block group the stability of the lower oxidation state increases. The ns^2 electrons are held tightly because the inner d and f electrons screen the nucleus poorly, so only the np electrons take part in bonding.

Group 13	B	Al	Ga	In	Tl
Oxidation states	III	III	III, I	III, I	III, **I** (more stable)
Group 14	C	Si	Ge	Sn	Pb
Oxidation states	IV	IV	IV, II	IV, II	IV, **II** (more stable)

$$\mathrm{Tl^+ > In^+ > Ga^+} \qquad \mathrm{Pb^{2+} > Sn^{2+} > Ge^{2+}}$$

Instability of the core. d-block ions do not reach an inert-gas core, and successive ionisation energies differ only moderately; hence several oxidation states are accessible.

Worked Example 3.1 Existence of Pb(IV) halides

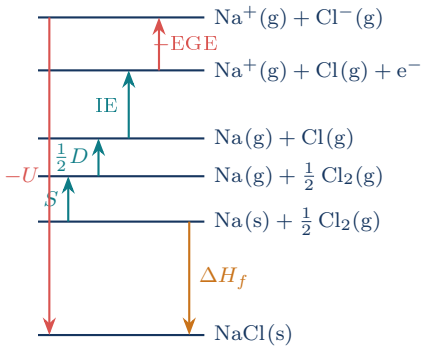
$PbCl_4$ is stable at room temperature, $PbBr_4$ exists but is unstable, and PbI_4 does not exist. Explain.
Solution. Because of the inert pair effect Pb(IV) is far less stable than Pb(II) and is a strong oxidant: $\mathrm{Pb(IV) + 2\,e^- \longrightarrow Pb(II)}$. Reducing power of the halide follows $\mathrm{I^- > Br^- > Cl^-}$, so

$$\mathrm{PbCl_4 \longrightarrow PbCl_2 + Cl_2} \qquad \mathrm{PbBr_4 \longrightarrow PbBr_2 + Br_2} \qquad \mathrm{PbI_4 \longrightarrow PbI_2 + I_2}$$

Pb(IV) and I^- cannot co-exist (I^- reduces Pb(IV) instantly), so PbI_4 does not exist.

3.2 Formation of Ionic Compound: Born–Haber Cycle

Spontaneity requires $\Delta G < 0$, with $\Delta G = \Delta H - T\Delta S$. A whole series of energy changes lies between the elements and the ionic crystal; Hess's law links them in the Born–Haber cycle.



S = enthalpy of sublimation of Na
D = bond dissociation enthalpy of Cl_2
IE = ionisation energy of Na(g)
EGE = electron gain enthalpy of Cl(g)
U = lattice energy of NaCl(s)
ΔH_f = enthalpy of formation

Figure 4. Born–Haber cycle for NaCl(s). Energies are not drawn to scale.

Formation of NaCl(s): $\mathrm{Na(s) + \tfrac{1}{2}\,Cl_2(g) \longrightarrow NaCl(s)}$.

Formula 3.1 Born–Haber Relation

$$\Delta H_f \;=\; S + \mathrm{IE} + \frac{D}{2} - \mathrm{EGE} - U$$

$$\boxed{\text{For } MX_2: \qquad \Delta H_f = S + (IE_1 + IE_2) + D - 2\,EGE - U}$$

Conditions

EGE and U are entered as positive energies *released*; S, IE and D are positive energies *absorbed*. $D/2$ is used for one mole of Cl atoms (diatomic halogen); for MX_2 the full D and 2 EGE appear. The cycle gives U indirectly because lattice energy cannot be measured directly.

Worked Example 3.2 Lattice energy of NaCl from the cycle

Given $S = 108$, IE$= 496$, $D = 244$, EGE$= 349$ and $\Delta H_f = -411\,\text{kJ mol}^{-1}$, find U.
Solution. Rearranging the Born–Haber relation,

$$\boxed{U = S + \text{IE} + \frac{D}{2} - \text{EGE} - \Delta H_f = 108 + 496 + \frac{244}{2} - 349 - (-411) = 787\,\text{kJ mol}^{-1}}$$

3.3 Factors Favouring Ionic Bonding

- **(a) Ionisation energy:** the lower the IE, the greater the tendency to form the cation: $Cs^+ > Rb^+ > K^+ > Na^+ > Li^+$ and $Na^+ > Mg^{2+} > Al^{3+}$.
- **(b) Electron affinity:** the higher the EA, the greater the tendency to form the anion: $Cl^- > F^- > Br^- > I^-$; $F^- > O^{2-} > N^{3-}$.
- **(c) Lattice energy:** the energy released when one mole of the ionic solid forms from its gaseous ions; the higher U, the more stable the compound.

Formula 3.2 Dependence of Lattice Energy

$$\boxed{U \;\propto\; \frac{z^+ z^-}{r^+ + r^-}}$$

Conditions

$z^{\pm}$ are the ionic charges and $r^+ + r^-$ is the interionic distance. At constant charge, U falls as cation size rises (LiCl > NaCl > KCl > RbCl > CsCl); at nearly constant size, U rises steeply with charge (NaCl < $MgCl_2$ < $AlCl_3$ on the ionic model, although $AlCl_3$ is in fact covalent – see Fajans' rules).

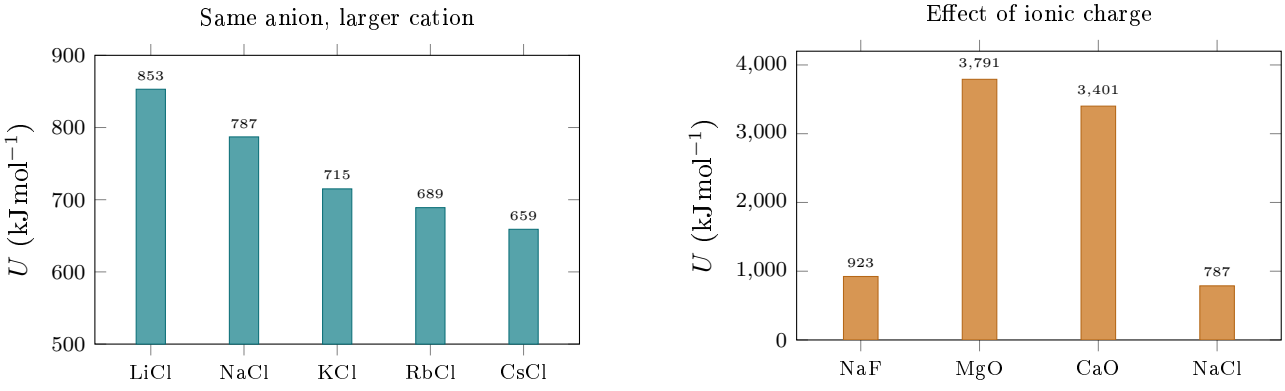


Figure 5. Lattice energies (standard Born–Haber values). Doubly charged MgO and CaO have lattice energies four to five times those of the singly charged halides.

3.4 Properties of Ionic Compounds

(a) **Physical state.** Hard, crystalline and brittle. On displacement of a layer, like-charged ions come face to face and repel, so the crystal cleaves.

(b) **Isomorphism.** Different ionic compounds with the same ionic geometry are *isomorphs*: NaF and MgO; $ZnSO_4 \cdot 7H_2O$ and $FeSO_4 \cdot 7H_2O$; all alums $M'_2SO_4{\cdot}M''_2(SO_4)_3 \cdot 24H_2O$.

(c) **Melting and boiling points.** High, owing to strong electrostatic attraction between oppositely charged ions.

(d) **Conductivity** depends on ionic mobility: no free ions in the solid state (bad conductor); free ions in the fused state or aqueous solution (good conductor). Solid < fused < aqueous solution.

(e) **Solubility** is the molar concentration of the saturated solution at a given temperature.

3.5 Factors Affecting Solubility

(a) Dielectric constant of the solvent

A solvent of high dielectric constant ε weakens the force between ions (Coulomb force $\propto 1/\varepsilon$) and so dissolves ionic compounds readily. Water has the highest value among common solvents.

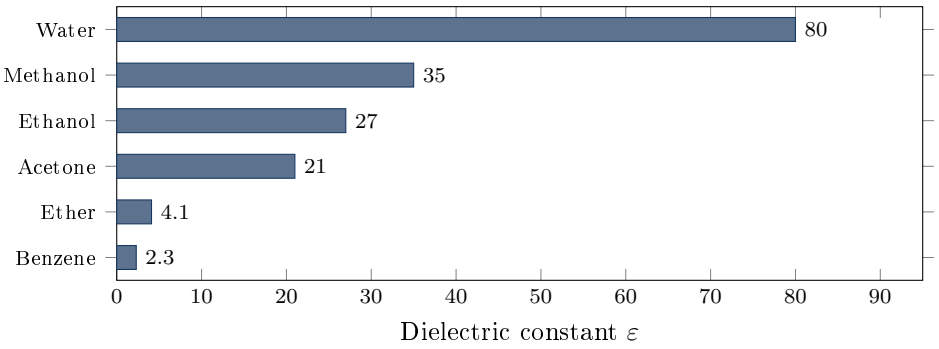


Figure 6. Dielectric constants: $H_2O > CH_3OH > C_2H_5OH > CH_3COCH_3 > CH_3OCH_3 > C_6H_6$.

Note

H_2SO_4 and H_2O_2 have high dielectric constants but are not used as solvents for salts because of their oxidising nature.

(b) Lattice energy and hydration energy

Derivation

Dissolution $MX(s) \longrightarrow M^+(aq) + X^-(aq)$ can be broken into (1) $MX(s) \rightarrow M^+(g)^+X^-(g)$, absorbing the lattice enthalpy U, and (2) hydration of the gaseous ions, releasing $|\Delta H_{hyd}|$. By Hess's law

$$\Delta H^\circ_{sol} = U - |\Delta H_{hyd}|, \qquad \Delta G^\circ = \Delta H^\circ - T\Delta S^\circ.$$

Formula 3.3 Solubility Criterion

$$\boxed{\Delta H^\circ_{sol} = U - |\Delta H_{hyd}| \ ; \qquad \text{soluble if} \quad |\Delta H_{hyd}| > U}$$

$$\boxed{U \propto \frac{1}{r^+ + r^-} \ , \qquad |\Delta H_{hyd}| \propto \frac{1}{r^+} + \frac{1}{r^-}}$$

Conditions

For ΔG° to be negative the hydration enthalpy must outweigh the lattice enthalpy (the entropy term $T\Delta S^\circ$ matters only when the two nearly cancel). If U decreases, solubility increases; if the hydration enthalpy decreases, solubility decreases. Down the s-block most salts become less soluble because hydration energy falls faster than lattice energy (for a large anion such as SO_4^{2-} or CO_3^{2-} the lattice energy hardly changes while the cation's hydration energy falls). With a small anion such as OH^- the lattice energy falls faster, so Group 2 hydroxide solubility rises down the group.

4 Transition from Ionic to Covalent Bond: Fajans' Rule

An ionic bond may have a degree of covalent character. When two ions approach, the cation attracts the outer electrons of the anion and repels its nucleus; the anion is *polarised* (distorted) and the electron cloud is partly shared.

- **Polarising power** of a cation: its ability to distort a neighbouring anion. It is measured by the ionic potential (charge density),

Formula 4.1 Ionic Potential

$$\phi = \frac{\text{charge on cation}}{\text{size of cation}} = \frac{z^+}{r^+}$$

- **Polarisability** of an anion: the ease with which it is distorted by the cation. Increase in polarisation increases covalent character.



Separate ions (ionic) Polarised anion (covalent character)

Figure 7. Polarisation of an anion by a cation, as in AgI.

4.1 Factors Influencing Ion Deformation (Increasing Covalent Character)

(i) **Large charge on the ions.** The greater the cation charge, the more strongly it attracts the anion's electrons: Al^{3+} polarises Cl^- more than Na^+, so $AlCl_3$ is covalent while NaCl, AlF_3 and AgF are ionic.

(ii) **Small cation and large anion.** A small cation exerts a stronger field and a large anion holds its outer electrons loosely, as in AgI.

(iii) **Cation with pseudo-inert gas configuration.** An 18-electron cation such as Ag^+ ($[Kr]4d^{10}$) polarises more strongly than an 8-electron cation like K^+ of similar size, because d-electrons shield the nuclear charge poorly. Thus AgI is more covalent than KI; Cu^+ and Hg_2^{2+} salts are covalent.

Formula 4.2 Fajans' Rule (Summary)

$$\text{Covalent character} \propto \underbrace{\frac{z^+}{r^+}}_{\text{polarising power}} \times \underbrace{z^- r^-}_{\text{polarisability}} \qquad (18e^- \text{ cation} > 8e^- \text{ cation})$$

Conditions

Qualitative ranking rule for comparing compounds with a *common* anion (or cation). Use it as a trend, not as a calculation.

4.2 Application of the Concept of Polarisation

(a) Covalent versus ionic character

- **Across a period** the cation charge rises and its size falls, so polarising power rises: $Na^+ < Mg^{2+} < Al^{3+} < Si^{4+}$ and covalent character increases.
- **Down a group** the size increases at fixed charge, so polarising power and covalent character decrease: $Li^+ > Na^+ > K^+ > Rb^+ > Cs^+$.

Cation	Li^+	Na^+	K^+	Rb^+	Cs^+	Mg^{2+}	Al^{3+}	Si^{4+}
Radius (pm)	60	95	133	148	169	65	50	41
$\phi = z/r$ ($\times$100 pm)	1.67	1.05	0.75	0.68	0.59	3.08	6.00	9.76

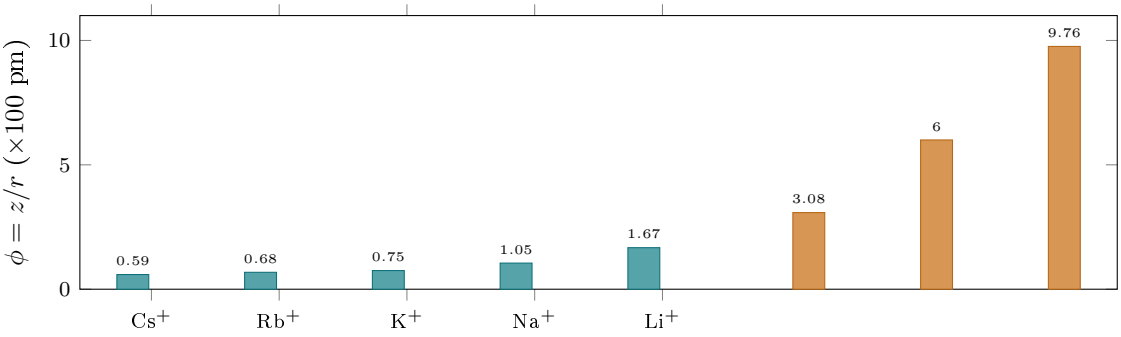


Figure 8. Ionic potential (polarising power) of cations. Covalent character in a chloride follows this order.

(b) Thermal stability of metal carbonates

Derivation

A small, highly charged cation M^{2+} polarises CO_3^{2-} and strengthens the M–O bond at the expense of the C–O bond, so the carbonate decomposes more easily:

$$MCO_3 \xrightarrow{\Delta} MO + CO_2$$

Thermal stability therefore varies inversely with the covalent character of the carbonate.

$$\text{Thermal stability of } MCO_3 \;\propto\; \frac{1}{\text{covalent character}} \;\propto\; \frac{r^+}{z^+}$$

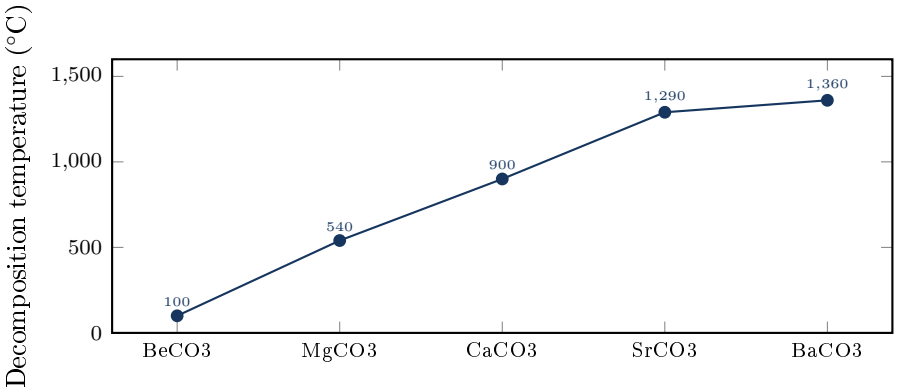


Figure 9. Approximate decomposition temperatures of Group 2 carbonates (representative values): stability increases down the group as cation size increases.

(c) Colour of compounds (colour density)

The greater the covalent character, the greater the colour intensity: $\underset{\text{white}}{AgCl} < \underset{\text{light yellow}}{AgBr} < \underset{\text{yellow}}{AgI}$ and $\underset{\text{white}}{SnO_2}$, $\underset{\text{yellow}}{SnS_2}$.

Worked Example 4.1 Most covalent oxide of manganese

Compare MnO, MnO_2, Mn_2O_7.
Solution. The oxidation state rises $Mn^{2+} < Mn^{4+} < Mn^{7+}$, and polarising power rises with it. Hence

$$\boxed{MnO < MnO_2 < Mn_2O_7 \quad \text{(covalent character)}}$$

Mn_2O_7 is a covalent liquid and an acidic oxide.

Worked Example 4.2 Thermal stability of $PbCO_3$, $CaCO_3$, $CdCO_3$

Solution. Ca^{2+} has an 8-electron shell; Cd^{2+} has a pseudo-inert 18-electron shell; Pb^{2+} has $(18+2)$ electrons. Polarising power: $Cd^{2+} > Pb^{2+} > Ca^{2+}$. Therefore

$$\boxed{CaCO_3 > PbCO_3 > CdCO_3 \quad \text{(thermal stability)}}$$

Worked Example 4.3 Covalent character of Group 2 chlorides

Solution. Down Group 2 the cation size increases and ϕ decreases:

$$\boxed{BeCl_2 > MgCl_2 > CaCl_2 > SrCl_2 \quad \text{(covalent character)}}$$

Part III — The Covalent Bond and Molecular Shape

5 Covalent Bond

A covalent bond is formed by the mutual sharing of electrons between two atoms of the same (or comparable) electronegativity so that each completes its octet (hydrogen completes a duplet). The shared pair has opposite spins and is localised between the two atoms.

Electrons shared	Bonded pairs	Bond	Examples
2	1	single (−)	H−H, H−O−H (two single bonds), NH_3 (three single bonds)
4	2	double (=)	O=O, C=O
6	3	triple (≡)	N≡N, HC≡CH

5.1 Characteristics of Covalent Compounds

(i) **Physical state.** Found as gases, liquids and solids. Molecules are separate in the gaseous state and associated (van der Waals forces, hydrogen bonding) in liquids and solids. As molecular size increases the state changes: F_2, Cl_2 (gas) → Br_2 (liquid) → I_2 (solid).

(ii) **Covalent solid versus molecular solid.** In a *covalent (network) solid* atoms are linked by covalent bonds into an infinite three-dimensional giant structure: diamond, graphite, AlN, SiC, SiO_2. In a *molecular solid* discrete molecules are held by van der Waals forces: solid I_2, dry ice (solid CO_2).

(iii) **Conductivity.** Mostly poor conductors. A few polar covalent compounds conduct through self-ionisation, e.g. $2\,H_2O \rightleftharpoons H_3O^+ + OH^-$ and $2\,NH_3 \rightleftharpoons NH_4^+ + NH_2^-$. Exceptions: graphite and aqueous HCl.

(iv) **Solubility.** Non-polar compounds dissolve in non-polar solvents through van der Waals interaction.

(v) **Isomerism.** The covalent bond is rigid and directional, hence isomerism (organic compounds).

(vi) **Reaction rate.** Reactions between covalent compounds are slow because existing bonds must be broken and new bonds formed.

5.2 Co-ordinate Bond

A co-ordinate (dative, donor–acceptor) bond is a covalent bond in which the shared pair is supplied by one atom only.

- **Donor:** octet complete, with at least one lone pair.
- **Acceptor:** deficient by at least one electron pair (empty orbital).

$H_3N{:}+BF_3 \longrightarrow H_3N\rightarrow BF_3$ $\quad NH_3+H^+ \longrightarrow NH_4^+$ $\quad Cu^{2+}+4\,NH_3 \longrightarrow [Cu(NH_3)_4]^{2+}$ $\quad N_2O:\ N\equiv N\rightarrow O$

Note

Compounds containing ionic, covalent and co-ordinate bonds together: NH_4Cl, $CuSO_4{\cdot}5H_2O$, $K_4[Fe(CN)_6]$, KNO_3.

5.3 Variable Valency in Covalent Bonds

Promotion rule. Elements with vacant orbitals in the outermost shell show variable covalency. Paired electrons are *excited* (promoted) to higher sub-shells of the same shell, giving the maximum number of unpaired electrons; the energy needed is the *promotion energy*. Maximum covalency is shown in the highest excited state.

Atom	State	3s	3p	3d	Covalency
P	Ground	↑↓	↑ ↑ ↑		3 (PCl_3)
	Excited	↑	↑ ↑ ↑	↑	5 (PCl_5)
S	Ground	↑↓	↑↓ ↑ ↑		2 (SF_2)
	1st excited	↑↓	↑ ↑ ↑	↑	4 (SF_4)
	2nd excited	↑	↑ ↑ ↑	↑ ↑	6 (SF_6)

Figure 10. Promotion of electrons within the third shell for P and S.

Iodine ($5s^25p^5$) has three lone pairs/paired sets and so passes through three excited states, reaching a maximum of seven unpaired electrons; its covalencies are $1, 3, 5, 7$ (likewise Cl in ClF, ClF_3, ClF_5 and F alone remains monovalent).

Covalency = number of unpaired electrons (ground or excited state)

Conditions

Excitation is within the *same* principal shell (promotion rule). Hence N ($2s^22p^3$, no $2d$) cannot be pentavalent, while P ($3s^23p^3$ with vacant 3d) can. S shows $2, 4, 6$; halogens (except F) $1, 3, 5, 7$.

6 Lewis Dot Structures or Lewis Symbols

Only the *valence* electrons take part in bonding; Lewis introduced dot notation for them. The number of dots around the symbol equals the number of valence electrons.



Figure 11. Lewis symbols for the second-period elements (first four dots placed singly, the rest paired).

Lewis symbols for anions place the extra electrons around the symbol and enclose it in brackets: $[:\ddot{F}:]^-$, $[:\ddot{O}:]^{2-}$.

Rules to draw the Lewis structure of an inorganic covalent species

1. **Central atom:** usually the least electronegative atom (H can never be central, its covalency being one); otherwise the atom that is fewer in number, or the one in the central position of the formula, e.g. $\underline{C}H_4$, $\underline{N}H_3$, $\underline{S}O_2$, $\underline{S}OCl_2$, $\underline{O}Cl_2$ (here the central atom is O, which is fewer in number although more electronegative), $(S\underline{C}N)^-$.
2. **Surrounding atoms** are those bonded directly to the central atom. Join them with single bonds.
3. Complete the octet of every surrounding atom; if single bonds are not enough, form double or triple bonds with the central atom.
4. Show the remaining lone pairs on central and surrounding atoms.
5. A central atom of the **second period** has at most 8 electrons (≤ 4 bonds); a central atom of the **third or lower period** may exceed an octet.
6. Verify the covalency of the central atom.
7. **For ions:** distribute the negative charge on surrounding atoms so that no surrounding atom has a complete octet before bonding. If a halogen (7 valence electrons) completes its octet by becoming uninegative, it is attached by a co-ordinate bond.

Formula 6.1 Total Valence Electrons (Counting Aid)

$$N_{\text{val}} = \sum_{\text{atoms}} V_i \; - \; q \qquad (q = \text{charge of the species, negative for anions})$$

Conditions

Use N_{val} to check that the number of dots plus two per bond equals the total. For NO_3^-: $5+3(6)+1=24$ electrons; for NH_4^+: $5+4-1=8$.

Worked Example 6.1 NO_3^- cannot have five bonds on nitrogen

A structure with N forming five covalent bonds is incorrect because N (period 2) cannot exceed eight electrons. The correct structure has one N=O double bond, one N–O^- single bond with an anionic O, and a further N–O^-; the negative charge is delocalised over the three O atoms (resonance), and nitrogen is sp^2 hybridised.

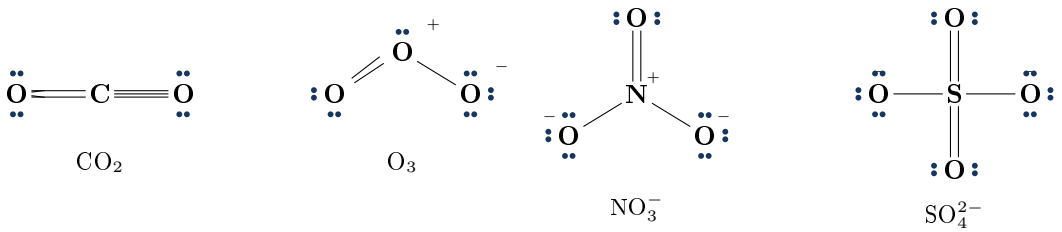


Figure 12. Lewis structures of CO_2, O_3, NO_3^- and SO_4^{2-} (one resonance form each; lone pairs shown as dot pairs; the third-period atom S in sulphate may expand its octet).

Species	Lewis description
NH_4^+	N with four N–H bonds ($8e^-$), one of which is co-ordinate; N carries +1
BF_4^-	B with four B–F bonds, one co-ordinate ($F^- \rightarrow B$); B carries −1
PO_4^{3-}, SO_3^{2-}	P (or S) at the centre; P with one P=O and three P–O^- (expanded octet)
PCl_6^-	P with six P–Cl bonds ($12e^-$, period 3 expands octet)
N_3^-	$N{=}N^+{=}N$ with terminal N bearing −1 each: $[N{=}N{=}N]^-$
SCN^-	$S{=}C{=}N^-$ (or $^-S{-}C{\equiv}N$)

6.1 Formal Charge

Lewis structures do not give true shapes, and in polyatomic ions the charge belongs to the ion as a whole. A *formal charge* can however be assigned to each atom by comparing the electrons it owns in the structure with its free-atom valence electrons.

Formula 6.2 Formal Charge

$$\text{F.C.} = V - L - \frac{1}{2}B$$

Conditions

V = valence electrons of the free atom; L = non-bonding (lone-pair) electrons; B = bonding (shared) electrons. The sum of formal charges equals the charge of the species. Formal charges are bookkeeping devices only; they do not imply real charge separation.

Species	Atom	Calculation	F.C.
O_3	central O (two bonds, one lone pair)	$6-2-\frac{1}{2}(6)$	$+1$
	O (double bond, two lone pairs)	$6-4-\frac{1}{2}(4)$	0
	O (single bond, three lone pairs)	$6-6-\frac{1}{2}(2)$	-1
SO_2	S	$6-2-\frac{1}{2}(6)$	$+1$
	O (single bond)	$6-6-\frac{1}{2}(2)$	-1
	O (double bond)	$6-4-\frac{1}{2}(4)$	0
CO	C (:C≡O:) / O	$4-2-\frac{1}{2}(6)$ / $6-2-\frac{1}{2}(6)$	-1 / $+1$
NH_4^+	N / each H	$5-0-\frac{1}{2}(8)$ / $1-0-\frac{1}{2}(2)$	$+1$ / 0
NO_2^-	N / O (double) / O (single)	$5-2-\frac{1}{2}(6)$; $6-4-\frac{1}{2}(4)$; $6-6-\frac{1}{2}(2)$	0 / 0 / -1

Selecting the best Lewis structure

1. Small formal charges are preferable to large ones; the lowest-energy structure has the smallest formal charges.
2. Negative formal charge should reside on the more electronegative atom.
3. Unlike charges close together are favoured over widely separated opposite charges.
4. Like charges on adjacent atoms are very unlikely.

Worked Example 6.2 Choosing between two Lewis structures

Compare $H_3C{-}C(=O){-}NH_2$-type structures: the form with all formal charges zero (C=O and N–H) is lower in energy than the charge-separated form in which O carries -1 and N carries $+1$, because the latter has opposite charges that must be separated and N^+ is electron-poor.

7 Valence Shell Electron Pair Repulsion (VSEPR) Theory

The Lewis picture cannot explain molecular shapes. Sidgwick and Powell (1940) proposed that the shape is decided by repulsion between valence-shell electron pairs; the theory was refined by Nyholm and Gillespie (1957).

Main postulates

1. The shape is determined by repulsion between *all* electron pairs (bond pairs and lone pairs) in the valence shell of the central atom.
2. A lone pair occupies more space than a bond pair, because it is attracted by one nucleus while a bond pair is shared by two. Hence

$$\text{l.p.–l.p.} \;>\; \text{l.p.–b.p.} \;>\; \text{b.p.–b.p.} \quad \text{(repulsion)}$$

Lone pairs therefore compress bond angles from the ideal value.

3. The repulsion between bond pairs depends on the electronegativity difference between the central atom and the surrounding atoms (see Bond Angle, §15.2).
4. Double bonds repel more than single bonds, and triple bonds more than double bonds.
5. The theory is especially accurate for compounds of p-block elements.

Pairs	b.p. + l.p.	Type	Electron geometry / shape	Example	Hybridisation
2	2+0	AB_2	linear, 180°	$BeCl_2$, CO_2	sp
3	3+0	AB_3	trigonal planar, 120°	BF_3, SO_3	sp^2
3	2+1	AB_2E	bent (V-shape)	SO_2, O_3, $SnCl_2$	sp^2
4	4+0	AB_4	tetrahedral, 109°28′	CH_4, NH_4^+	sp^3
4	3+1	AB_3E	trigonal pyramidal	NH_3, PCl_3	sp^3
4	2+2	AB_2E_2	bent	H_2O, OF_2	sp^3
5	5+0	AB_5	trigonal bipyramidal	PCl_5	sp^3d
5	4+1	AB_4E	see-saw	SF_4	sp^3d
5	3+2	AB_3E_2	T-shaped	ClF_3	sp^3d
5	2+3	AB_2E_3	linear	XeF_2, I_3^-	sp^3d
6	6+0	AB_6	octahedral, 90°	SF_6	sp^3d^2
6	5+1	AB_5E	square pyramidal	IF_5, BrF_5	sp^3d^2
6	4+2	AB_4E_2	square planar	XeF_4, ICl_4^-	sp^3d^2
7	7+0	AB_7	pentagonal bipyramidal	IF_7	sp^3d^3
7	6+1	AB_6E	distorted octahedral	XeF_6	sp^3d^3

Common Trap

In trigonal bipyramidal geometries lone pairs occupy *equatorial* positions (fewer 90° interactions), which gives see-saw SF_4, T-shaped ClF_3 and linear XeF_2. Shape names refer to the arrangement of atoms only, never of lone pairs.

Part IV — Quantum-Mechanical Theories of the Covalent Bond

8 Wave Mechanical Concept of Covalent Bonding

8.1 Modern Concept of Covalent Bond (VBT)

The Lewis approach helps write structures but does not explain *how* a bond forms. It also gives no reason for the difference in bond enthalpy and bond length between H_2 (436 kJ mol^{-1}, 74 pm) and F_2 (158 kJ mol^{-1}, 144 pm) although each has a single shared pair, and it says nothing of the shapes of polyatomic molecules. VSEPR predicts geometry but does not explain it. Two theories based on quantum mechanics remove these limitations:

- **Valence Bond (VB) theory** – bonds from overlap of atomic orbitals (Heitler–London, 1927; extended by Pauling);
- **Molecular Orbital (MO) theory** – electrons delocalised over the whole molecule (Hund and Mulliken, 1932).

9 Valence Bond Theory (VBT)

VBT was introduced by Heitler and London (1927) and developed by Pauling and others. It rests on atomic orbitals, electronic configurations, the overlap criterion and hybridisation.

Formation of the H_2 molecule

As two H atoms approach, three interactions operate: (i) electron–electron repulsion, (ii) electron–proton attraction, (iii) proton–proton repulsion. Experimentally the attractions outweigh the repulsions, so the potential energy falls until a minimum is reached at the point where attraction and repulsion balance; the atoms are then bonded.

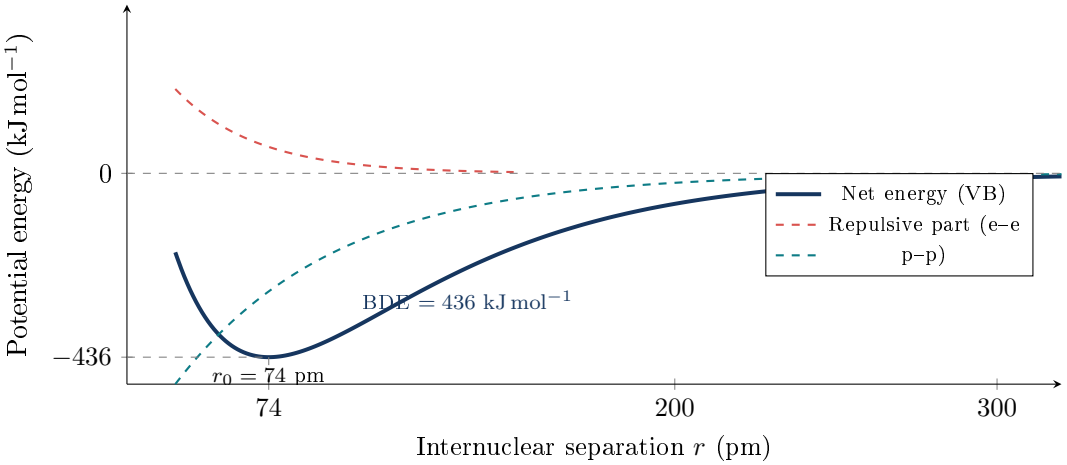


Figure 13. Potential energy of two hydrogen atoms versus internuclear separation. The curves for repulsive and attractive contributions are schematic; the net curve uses a Morse function.

Term	Definition
Bond length	Internuclear distance at minimum potential energy (74 pm for H_2)
Bond dissociation energy (BDE)	Energy required to break one mole of a given bond in the gaseous state; numerically equal to the energy released when the bond forms. For H_2: $H_2(g) + 436\,kJ \longrightarrow 2\,H(g)$

10 Orbital Overlap Concept

In H_2 the energy is a minimum when the two atomic orbitals undergo partial interpenetration. This merging is *overlapping*; it pairs the electrons (of opposite spin) and the extent of overlap decides the bond strength.

Formula 10.1 Overlap Criterion

$$\boxed{\text{Bond strength } \propto \text{ extent of orbital overlap}}$$

Conditions

Valid for bonds between orbitals of comparable energy and proper symmetry. Overlap is favoured for orbitals directed towards each other, which is why a bond is *directional*: the geometry of bonds in CH_4, NH_3 and H_2O is explained by overlap together with hybridisation.

Positive, negative and zero overlap

Overlap may be *positive* (lobes of the same sign), *negative* (opposite signs, destructive) or *zero* (equal positive and negative contributions cancel, e.g. an s orbital approaching a p orbital perpendicular to the bond axis).

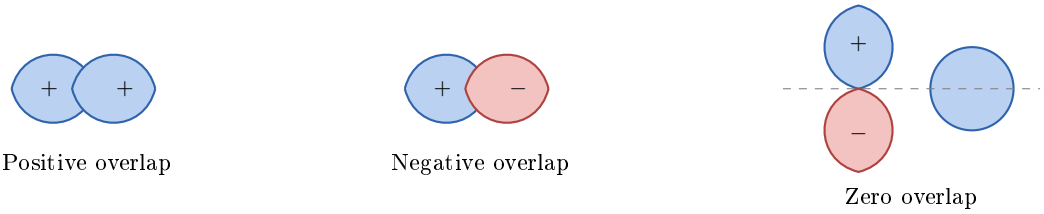


Figure 14. Positive, negative and zero overlap. In the third picture an s orbital approaches a p orbital sideways; the in-phase and out-of-phase overlaps cancel.

10.1 Types of Overlapping and Nature of Covalent Bonds

(i) **Sigma (σ) bond.** Formed by end-to-end (head-on, axial) overlap along the internuclear axis: s–s, s–p, p–p (axial).

(ii) **Pi (π) bond.** Formed by sideways overlap in which the orbital axes are parallel and perpendicular to the internuclear axis; two charge clouds lie above and below the plane of the atoms.

(iii) **Delta (δ) bond.** Formed by overlap of four lobes of two d orbitals (d_{xy}–d_{xy}, etc.); found in some transition-metal compounds.

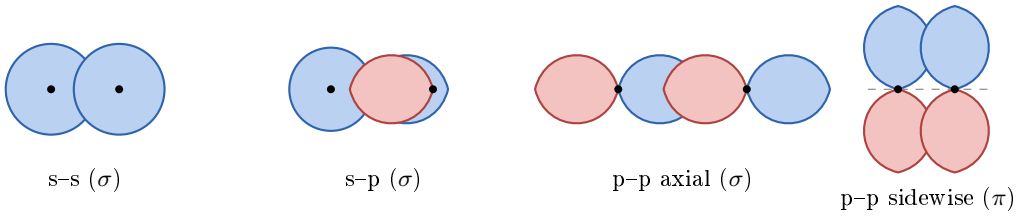


Figure 15. Types of overlap. The σ bonds are cylindrically symmetrical about the internuclear axis; the π bond has a nodal plane containing the axis.

	Sigma (σ) bond	Pi (π) bond
1	Axial (end-to-end) overlap	Sideways overlap
2	s–s, s–p, p–p overlap	p–p (also p–d, d–d) overlap
3	Stronger: larger extent of overlap	Weaker: smaller extent of overlap
4	Symmetrical about the internuclear axis; charge density on the axis	Charge density above and below the axis; nodal plane along the axis
5	Free rotation about the bond	No free rotation
6	Can exist alone or with π bonds	Always accompanies a σ bond
7	Pure or hybrid orbitals are involved	Hybrid orbitals are never involved

Formula 10.2 Counting σ and π Bonds

$$\boxed{\text{single bond} = 1\sigma\ ; \qquad \text{double bond} = 1\sigma + 1\pi\ ; \qquad \text{triple bond} = 1\sigma + 2\pi}$$

Worked Example 10.1 σ and π bonds in tetracyanomethane, $C(CN)_4$

Solution. The molecule has four C−C single bonds and four C≡N triple bonds:

$$\boxed{\sigma = 4 \times 1 + 4 \times 1 = 8\ , \qquad \pi = 4 \times 2 = 8}$$

Worked Example 10.2 Which combination does not form a covalent bond (x = internuclear axis)?

Options: (A) s + p_y, (B) p_y + p_y, (C) p_z + p_z, (D) d_{xy} + d_{xy}.
Solution. The s orbital is symmetric in all directions, but p_y is perpendicular to the x axis: the overlaps

cancel (zero overlap). Hence **(A)** gives no bond; (B) and (C) form π bonds; (D) forms a δ bond.

Worked Example 10.3 Which p–d overlap forms a π bond?

Solution. For a π overlap the orbitals must be parallel and both perpendicular to the bond axis with the same lobe symmetry. For $p_z + d_{yz}$ along the y axis, both have lobes along z and overlap sideways: a pπ–dπ bond, answer **(D)**. Combinations such as $p_x + d_{yz}$ (x axis) or $d_{x^2-y^2} + d_{x^2-y^2}$ (z axis) give zero overlap.

Advantages and disadvantages of VBT

Advantages	Disadvantages
Explains bond length and bond strength; gives the relation between the extent of overlap and bond energy.	Predicts 90° angles for CH_4 (p–p overlap), NH_3 and H_2O; observed: 109°28′, 107°, 104.5°.
Accounts for the shape of molecules and for covalencies that do not match the number of half-filled orbitals in the ground state.	Cannot explain the paramagnetism of O_2 (needs MO theory).
Explains stability of octet-exception molecules (BF_3, $AlCl_3$, PCl_5, SF_6).	Needs hybridisation (Pauling) to explain geometry.

11 Hybridisation Theory

Introduced by Pauling and Slater to explain the equivalence of covalent bonds in a molecule. In $BeCl_2$ without hybridisation, one Be–Cl bond would be s–p and the other p–p, with different strengths and lengths. Experimentally both are identical, which is explained if the 2s and one 2p orbital of Be mix.

Definition

Hybridisation is the mixing of atomic orbitals of different shapes and approximately equal energy, and their redistribution of energy, to give an equal number of new orbitals of the same shape and same energy, called hybrid orbitals.

11.1 Characteristics of Hybridisation

(a) Orbitals, not electrons, are mixed: filled, half-filled and empty orbitals can all take part.

(b) The number of hybrid orbitals equals the number of atomic orbitals mixed.

(c) Each hybrid orbital has a large lobe (used for bonding) and a small lobe.

(d) The number of hybrid orbitals on the central atom equals the number of σ bonds plus lone pairs.

(e) One element can show several hybridisation states (C: sp, sp^2, sp^3).

(f) Hybrid orbitals are more directional than pure orbitals and therefore form stronger σ bonds.

11.2 Determination of the Hybridisation State

Method I. Count σ bonds and lone pairs on the central atom:

$$\boxed{\text{Number of hybrid orbitals } (H) = \text{number of } \sigma \text{ bonds} + \text{number of lone pairs}}$$

Method II.

Formula 11.1 Hybrid-Orbital Count (Method II)

$$\boxed{H = \frac{1}{2}\Big[V + M - C + A\Big]}$$

Conditions

V = valence electrons of the central atom; M = number of *monovalent* surrounding atoms (H, halogens); C = charge on a cation; A = charge on an anion. Divalent atoms such as O in oxo-ligands do not enter M (only the charge is added). $H = 2, 3, 4, 5, 6, 7$ corresponds to sp, sp^2, sp^3, sp^3d, sp^3d^2, sp^3d^3 respectively.

Species	Calculation	H	Hybridisation	Shape
NH_4^+	$\frac{1}{2}[5+4-1]$	4	sp^3	tetrahedral
SF_4	$\frac{1}{2}[6+4]$	5	sp^3d	see-saw
SO_4^{2-}	$\frac{1}{2}[6+0+2]$	4	sp^3	tetrahedral
NO_3^-	$\frac{1}{2}[5+0+1]$	3	sp^2	trigonal planar
XeF_4	$\frac{1}{2}[8+4]$	6	sp^3d^2	square planar
ICl_4^-	$\frac{1}{2}[7+4+1]$	6	sp^3d^2	square planar
IF_7	$\frac{1}{2}[7+7]$	7	sp^3d^3	pentagonal bipyramidal

11.3 Types of Hybridisation

Type	Orbitals mixed	s-char.	Angle(s)	Shape	Examples
sp	s + p	50%	180°	linear	$BeCl_2$, CO_2, C_2H_2, NO_2^+
sp^2	s + 2p	33.3%	120°	trigonal planar	BCl_3, BF_3, CO_3^{2-}, NO_3^-
sp^3	s + 3p	25%	109°28′	tetrahedral	CH_4, NH_4^+, SO_4^{2-}
sp^3d	s + 3p + d_{z^2}	20%	120°, 90°	trigonal bipyramidal	PCl_5
sp^3d^2	s + 3p + $d_{x^2-y^2}$, d_{z^2}	16.7%	90°	octahedral	SF_6, AlF_6^{3-}, PF_6^-, XeF_4, ICl_5
sp^3d^3	s + 3p + d_{xy}, $d_{x^2-y^2}$, d_{z^2}	14.3%	72°, 90°	pentagonal bipyramidal	IF_7, XeF_6

Formula 11.2 s-Character of a Hybrid Orbital

$$\%\, s = \frac{1}{H} \times 100\ , \qquad \%\, p = 100 - \%\, s$$

Conditions

Applies to hybrid sets with equal contribution of each atomic orbital (sp, sp^2, sp^3 ...). In molecules with lone pairs or unlike substituents the hybrid orbitals become non-equivalent (see Bent's rule).

(i) sp hybridisation: $BeCl_2$

Be state	Configuration	2s	2p
Ground	$2s^2$	↑↓	☐ ☐ ☐
Excited	$2s^1 2p^1$	↑	↑ ☐ ☐
Hybridised	two sp	↑ ↑	☐ ☐

(ii) sp^2 hybridisation: BCl_3 (iii) sp^3 hybridisation: CH_4

Atom	State	2s	2p
B	Ground		
B	Excited		
B	Hybridised (3 sp^2)		
C	Ground		
C	Excited		
C	Hybridised (4 sp^3)		

(iv) sp^3d hybridisation: PCl_5 (v) sp^3d^2: SF_6

In PCl_5 the d_{z^2} orbital hybridises with the s and three p orbitals. Three equatorial P−Cl bonds lie at 120° and two axial bonds are perpendicular to the plane. The axial bonds are longer than the equatorial ones because each axial bond pair experiences three 90° repulsions with the equatorial pairs, whereas an equatorial pair experiences only two. In SF_6 (second excited state of S), two d orbitals ($d_{x^2-y^2}$ and d_{z^2}) hybridise to give six equivalent sp^3d^2 orbitals directed to the corners of a regular octahedron (90°).

Atom	State	3s	3p	3d
P	Excited (PCl_5)			
P	sp^3d hybrid			
S	2nd excited (SF_6)			
S	sp^3d^2 hybrid			

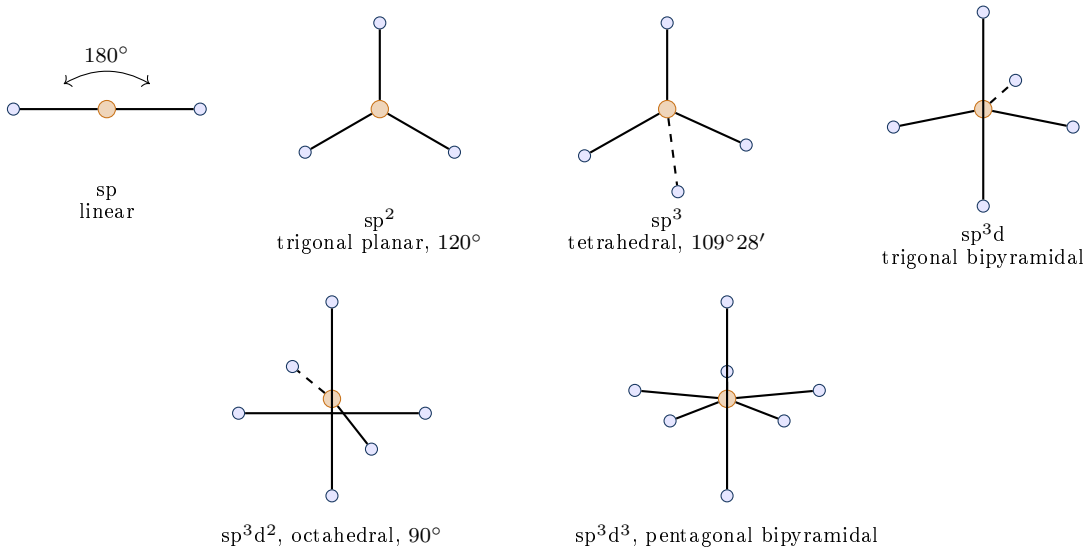


Figure 16. Geometries associated with the hybridisation types (ligands drawn as small circles; dashed lines are bonds directed away from the viewer).

Worked Example 11.1 sp^2 species

Which of CO_3^{2-}, NO_3^- and BF_3 are sp^2? $H = \frac{1}{2}[4+2] = 3$; $\frac{1}{2}[5+1] = 3$; $\frac{1}{2}[3+3] = 3$. **All three** are sp^2.

Worked Example 11.2 Electronegativity of hybridised carbon

Arrange the electronegativity of sp^3, sp^2 and sp carbon.
Solution. Electronegativity increases with s-character (s electrons are nearer the nucleus):

$$\boxed{\chi(\mathrm{sp}) > \chi(\mathrm{sp}^2) > \chi(\mathrm{sp}^3) \qquad (50\% > 33\% > 25\%\ \text{s-character})}$$

JEE Pointer

The ten-second recipe: draw the Lewis structure, count σ bonds + lone pairs on the central atom, then read off the hybridisation from $2 \to \mathrm{sp}$, $3 \to \mathrm{sp}^2$, $4 \to \mathrm{sp}^3$, $5 \to \mathrm{sp}^3\mathrm{d}$, $6 \to \mathrm{sp}^3\mathrm{d}^2$, $7 \to \mathrm{sp}^3\mathrm{d}^3$. Never count π bonds.

12 Hybridisation in Solid State

Several covalent halides adopt *ionic* lattices in the solid state; the hybridisation of the cationic and anionic parts then differs from that in the gas phase.

Species (solid)	Cationic part	Anionic part
PCl_5	PCl_4^+ (sp^3)	PCl_6^- (sp^3d^2)
PBr_5	PBr_4^+ (sp^3)	Br^-
XeF_6	XeF_5^+ (sp^3d^2)	F^-
N_2O_5	NO_2^+ (sp)	NO_3^- (sp^2)
I_2Cl_6 (liquid)	ICl_2^+ (sp^3)	ICl_4^- (sp^3d^2)
Cl_2O_6	ClO_2^+ (sp^2)	ClO_4^- (sp^3)
$(BeCl_2)_n$	chain polymer: each Be is sp^3 with two bridging Cl	

13 Bent's Rule

Formula 13.1 Bent's Rule

(i) A lone pair prefers the hybrid orbital with *greater* s-character.
(ii) A more electronegative substituent prefers the hybrid orbital with *smaller* s-character.

$$\boxed{\text{lone pair } \to \uparrow \%\, s\ ; \qquad \text{more electronegative atom/group } \to \downarrow \%\, s}$$

Reasoning

An s electron lies closer to the nucleus, so a lone pair in an s-rich orbital is more stabilised. A more electronegative atom draws the shared pair towards itself; by overlapping with a p-rich (more distant) hybrid it leaves the s-rich orbitals for bonds to less electronegative atoms and lone pairs, which raises the stability of the system.

Worked Example 13.1 C−C bond length in C_2H_6 and C_2Cl_6

Both carbons are sp^3 with no lone pair, but the four hybrid orbitals of each C are no longer equivalent.
Solution. In C_2Cl_6 three electronegative Cl atoms take p-rich orbitals on each carbon, leaving an s-rich hybrid for the C−C bond; in C_2H_6 H is less electronegative, so the C−C bond has comparatively more p-character. More s-character gives a shorter bond:

$$\boxed{d_{C-C}(C_2H_6) > d_{C-C}(C_2Cl_6) \quad \text{(Bent's-rule prediction)}}$$

Worked Example 13.2 Structure of PCl_3F_2

Solution. The axial positions of the trigonal bipyramid have less s-character (they are formed from p and d character), so the more electronegative F atoms occupy the **axial** sites and the Cl atoms the equatorial sites: F_{ax}, $Cl_{eq} \times 3$, F_{ax}.

14 Drago Generalisation

Formula 14.1 Drago Generalisation

$$\text{no hybridisation} \implies \text{bond angle} \approx 90^\circ$$

Conditions

Applies to molecules that (i) belong to the *third or lower period*, (ii) contain at least one *lone pair* on the central atom and (iii) have surrounding atoms of electronegativity ≤ 2.5 (e.g. H). Bonding then occurs through nearly pure p orbitals and the lone pair resides in an almost pure s orbital.

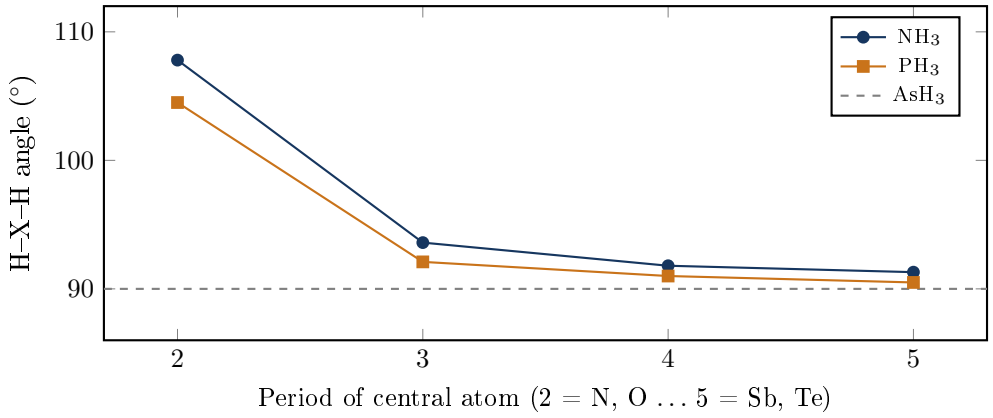


Figure 17. Bond angles of group 15 and group 16 hydrides approach 90° beyond the second period, as required by the Drago generalisation.

Group 15	Angle	Group 16	Angle
NH_3	107°48′	H_2O	104°28′
PH_3	93°36′	H_2S	92°
AsH_3	91°48′	H_2Se	91°
SbH_3	91°18′	H_2Te	90.5°

$$H_2O > H_2S > H_2Se > H_2Te \qquad NH_3 > PH_3 > AsH_3 > SbH_3$$

Part V — Bond Parameters and Molecular Properties

15 Bond Parameters

15.1 Bond Length

Bond length is the internuclear distance between two singly bonded atoms (at minimum potential energy).

Formula 15.1 Bond Length from Covalent Radii

$$d_{A-A} = r_A + r_A \quad (\chi_A = \chi_B) \qquad d_{A-B} < r_A + r_B \quad (\chi_A \neq \chi_B)$$

$$d_{A-B} = r_A + r_B - 0.09\,|\chi_A - \chi_B| \qquad (d,\ r \text{ in Å})$$

Conditions

The first relation holds for non-polar single bonds; the second (Schomaker–Stevenson) for polar bonds, where ionic character shortens the bond. Radii shrink for multiple bonds (C−C 77, C=C 67, C≡C 60 pm).

Worked Example 15.1 Bond length of H−Cl

$r_{H} = 0.37$ Å, $r_{Cl} = 0.99$ Å, $\chi_{H} = 2.1$, $\chi_{Cl} = 3.0$.

Solution.

$$\boxed{d_{H-Cl} = 0.37 + 0.99 - 0.09(3.0 - 2.1) = 1.36 - 0.081 = 1.28 \text{ Å}}$$

(experimental 1.27 Å).

Factors affecting bond length

(i) **Size of the atoms.** Bond length increases with size: HI > HBr > HCl > HF; $H_2Te > H_2Se > H_2S > H_2O$.

(ii) **Hybridisation of the bonded atoms.** Greater s-character gives a shorter bond (C–C and C–H).

(iii) **Resonance/delocalisation** gives bond lengths intermediate between single and double (e.g. the two O−O bonds in O_3 are both 128 pm; N−O in NO_2).

(iv) **Bond order.** Higher bond order, shorter bond (Illustration below).

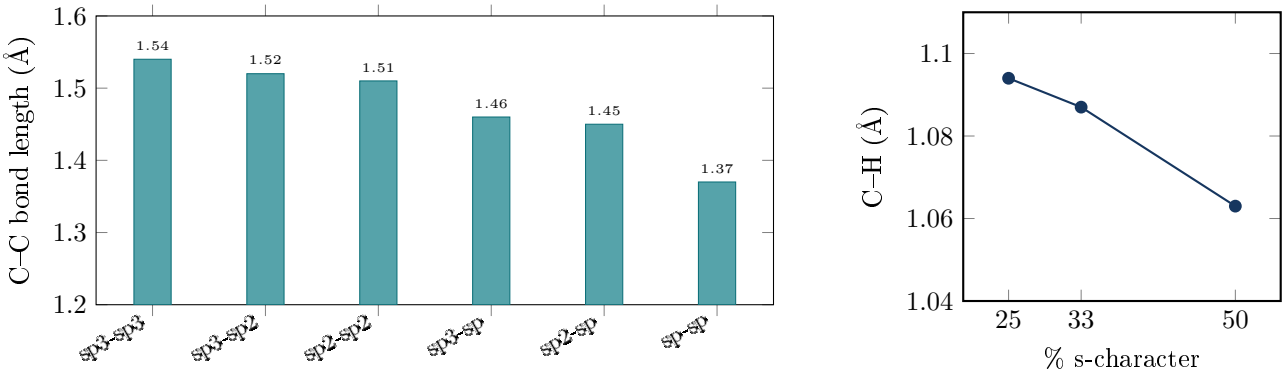


Figure 18. Left: C−C single-bond length decreases as the s-character of the bonded carbons increases. Right: C−H length falls from sp^3 (ethane) to sp (ethyne).

Bond	Type	d (pm)	Bond	Type	d (pm)	Molecule	d (pm)
O−H	single	96	C=O	double	121	H_2	74
C−H	single	107	N=O	double	122	F_2	144
N−O	single	136	C=C	double	133	Cl_2	199
C−O	single	143	C=N	double	138	Br_2	228
C−N	single	143	C≡N	triple	116	I_2	267
C−C	single	154	C≡C	triple	120	N_2	109
						O_2	121

Atom	H	C	N	O	F	Cl	Br	I	P	S
r_{cov} (pm)	37	77	74	66	64	99	114	133	110	104
Double bond	–	67	65	57	–	–	–	–	–	95
Triple bond	–	60	55	–	–	–	–	–	–	–

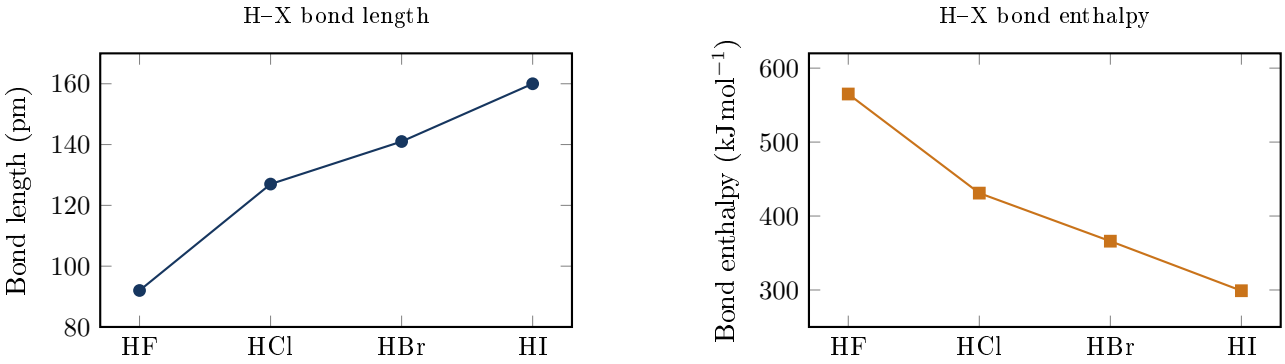


Figure 19. Hydrogen halides: the bond lengthens and weakens down the group.

15.2 Bond Angle

The angle between two adjacent bonds at an atom.

Factor	Effect	Examples
(i) Hybridisation of the central atom	Angle decreases with decreasing s-character	BeH_2 (sp) 180° > BH_3 (sp^2) 120° > CH_4 (sp^3) 109°28′
(ii) Lone pairs on the central atom	Each lone pair reduces the angle (l.p.–l.p. > l.p.–b.p. > b.p.–b.p.)	CH_4 109°28′ > NH_3 107° > H_2O 105°
(iii) Electronegativity of the central atom	Higher χ of central atom pulls bond pairs closer, so repulsion and angle rise	$H_2O > H_2S$; $NH_3 > PH_3$
(iv) Size of the terminal atoms	Larger terminal atoms increase the angle (steric crowding)	$OF_2 < Cl_2O < Br_2O$; $PF_3 < PCl_3 < PBr_3$

Note

When the electronegativity of the *surrounding* atom rises, bond pairs are drawn away from the central atom and the angle decreases (e.g. NH_3 107° > NF_3 102°), the reverse of the central-atom effect. For ≥ third-period central atoms with $\chi \leq 2.5$ substituents, apply the Drago generalisation instead.

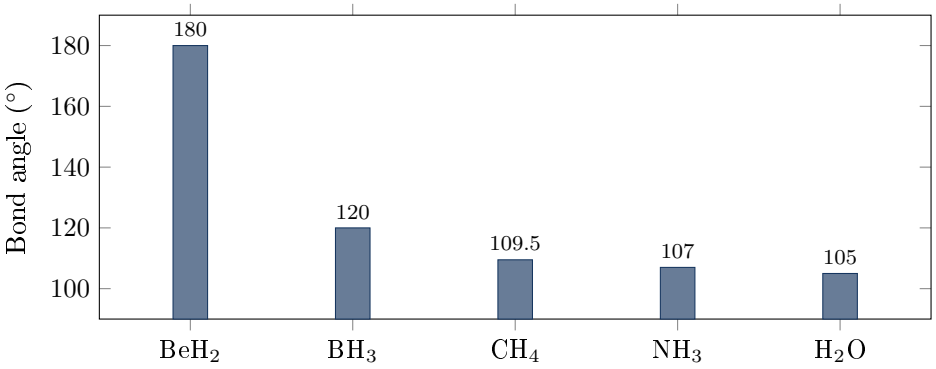


Figure 20. Bond angle versus hybridisation and number of lone pairs.

15.3 Bond Energy

The energy required to break one mole of a particular bond in the gaseous state (kJ mol^{-1} or kcal mol^{-1}).

Factor	Effect and examples
(a) Electronegativity difference	Larger $\Delta\chi$ gives higher ionic contribution and stronger bond: H−F > H−Cl > H−Br > H−I
(b) Bond order	Bond energy $\propto$ bond order: C−C < C=C < C≡C
(c) Hybridisation	Bond energy $\propto$ s-character: C_{sp^3}−H < C_{sp^2}−H < C_{sp}−H
(d) Atomic size	Small atoms overlap better: Cl−Cl > Br−Br > I−I
(e) Extent of overlap	Greater overlap, stronger bond
(f) Lone-pair repulsion on bonded atoms	Bond energy $\propto$ 1/(number of lone pairs on the bonded atoms): C−C > N−N > O−O > F−F (0, 1, 2, 3 lone pairs)

Common Trap

The bond energy order Cl−Cl > F−F is a classical exception to the size rule (rule (a) vs (d)): F is so small that lone-pair repulsion across the short F−F bond weakens it. Likewise S−S > O−O and P−P > N−N.

Formula 15.2 Bond Order, Bond Length and Bond Energy

$$\text{Bond order} \;\propto\; \text{bond dissociation energy} \;\propto\; \frac{1}{\text{bond length}}$$

Conditions

Valid for comparisons between bonds of the same two atoms (C−C, C=C, C≡C; or N_2, N_2^+, N_2^-).

16 Typical Concepts in Bonding

16.1 Odd Electronic Species

Species	Hybridisation	Shape	Remarks
NO_2	sp^2	bent	Both N−O bonds are intermediate between single and double; the odd electron occupies one sp^2 orbital. Paramagnetic brown gas; dimerises on cooling to -11°C to colourless diamagnetic N_2O_4.
ClO_2	sp^3-type	bent	Cl−O bond length lies between single and double; the odd electron is delocalised (in a d orbital of Cl), so dimerisation tendency is much smaller than for NO_2.
ClO_3	sp^3	pyramidal ($\approx 119^\circ$ per notes)	Readily dimerises to Cl_2O_6.
$\cdot CH_3$	sp^2	planar, 120°	Odd electron in the unhybridised p_z; dimerises to C_2H_6.
$\cdot CF_3$	sp^3	pyramidal, $109^\circ 28'$	Odd electron in an sp^3 orbital; electronegative F increases p-character in its bonding orbitals so carbon retains sp^3 (Bent's rule).

16.2 Molecules that Do Not Exist

Exists	Does not exist	Reason
SF_4, SF_6, PF_5	OF_4, OF_6, NF_5	N and O have no vacant 2d orbitals: no octet expansion
PCl_5, $TeCl_6$	PI_5 (vapour), SCl_6	Steric crowding of the large halogen atoms around a small central atom; Te > S in size
SF_6, PF_5, XeF_6, XeF_4, XeF_2	SH_6, PH_5, XeH_6, XeH_4, XeH_2	F is highly electronegative and contracts d orbitals so that sp^3d^n hybridisation is possible; H is not electronegative enough

Worked Example 16.1 Bond order and bond properties

Higher bond order means greater (A) bond dissociation energy, (B) covalent character, (C) bond length, (D) paramagnetism?
Solution. Bond order $\propto$ bond dissociation energy $\propto 1/$bond length. Answer **(A)**.

Worked Example 16.2 Non-existent due to steric reasons

Among F_3^-, Cl_3^-, Br_3^-, I_3^- which is non-existent for steric reasons (JEE convention)?
Solution. F_3^- is excluded for a different reason (no vacant d orbital in F). By the convention of the notes, Cl_3^- is the steric exception, whereas Br_3^- and I_3^- exist. Answer **(B)**.

Worked Example 16.3 Non-existing pair

(A) XeH_2, XeF_4 (B) PH_5, PCl_5 (C) PF_5, XeF_2 (D) XeH_4, SH_6
Solution. A hydride cannot cause d-orbital contraction, so XeH_4 and SH_6 do not exist. Answer **(D)**.

17 Dipole Moments and Molecular Polarity

The polarity of a bond is measured by its dipole moment.

Formula 17.1 Dipole Moment

$$\boxed{\mu = q \times d}$$

$$\boxed{1\ \text{D} = 10^{-18}\ \text{esu cm} = 3.336 \times 10^{-30}\ \text{C m}}$$

Conditions

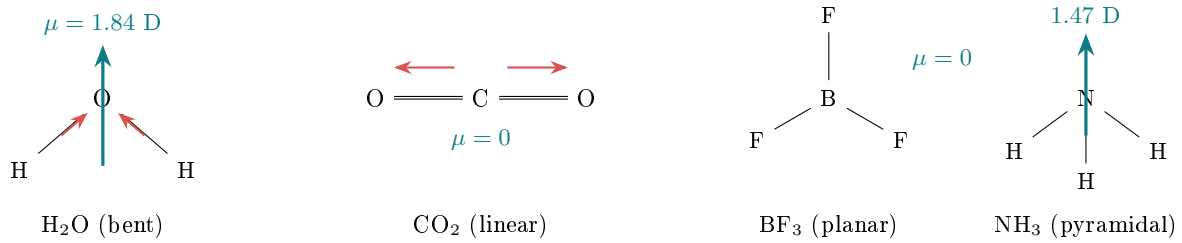
q is the magnitude of charge separated ($\approx 10^{-10}$ esu; unit charge $= 4.8 \times 10^{-10}$ esu $= 1.602 \times 10^{-19}$ C) and d the distance between the centres ($\approx 10^{-8}$ cm). μ is a vector pointing from the positive to the negative end. A polar *bond* does not make a polar *molecule*: the molecular moment is the vector sum of bond moments.

Formula 17.2 Resultant of Two Bond Moments

$$\boxed{\mu_{\text{res}} = \sqrt{\mu_1^2 + \mu_2^2 + 2\mu_1\mu_2\cos\theta}}$$

Conditions

θ is the angle between the two moment vectors (the bond angle when both point away from, or towards, the central atom). For $\theta = 180^\circ$ and $\mu_1 = \mu_2$ the resultant is zero (CO_2); for $\theta = 0^\circ$ it is $\mu_1 + \mu_2$.



H_2O (bent) CO_2 (linear) BF_3 (planar) NH_3 (pyramidal)

Figure 21. Vector addition of bond moments. The resultant arrow points towards the negative end.

Worked Example 17.1 Dipole moment of water from bond moments

Each O−H bond moment is 1.5 D and $\angle HOH = 104.5°$.

Solution. The two moments make an angle $\theta = 104.5°$; $\mu_1 = \mu_2 = 1.5$ D,

$$\boxed{\mu = \sqrt{2(1.5)^2\,[1 + \cos 104.5°]} = 2(1.5)\cos 52.25° = 1.84 \text{ D}}$$

which agrees with the measured value 1.84–1.85 D.

Methane series

$$\boxed{\mu:\ CH_3Cl > CH_2Cl_2 > CHCl_3 > CCl_4 = CH_4\ (=0)}$$

17.1 Application of Dipole Moment Measurements

(i) Inorganic substances

Class	Zero dipole moment (symmetrical)	Non-zero (unsymmetrical)
Monoatomic	He, Ne	–
Homodiatomic	H_2, Cl_2, N_2	–
Triatomic	CO_2, CS_2, $HgCl_2$ (linear)	H_2O (1.84 D), SO_2 (1.63 D): bent
Tetra-atomic	BF_3, BCl_3, BBr_3, CO_3^{2-}, NO_3^- (planar)	NH_3, PCl_3, $AsCl_3$, PH_3, AsH_3, H_3O^+ (pyramidal)
Others	CH_4, CCl_4, SF_6, XeF_4, PCl_5	SF_4, $CHCl_3$, CH_3Cl, PCl_2F_3

(ii) Organic substances

- CH_4 and CCl_4 have zero moment: symmetrical tetrahedra. Benzene ($\mu = 0$): symmetrical planar hexagon.
- The **trans** isomer of a symmetrical alkene (1,2-dibromoethene) has $\mu = 0$, the **cis** isomer has a definite moment (1.4 D); dipole moments therefore distinguish geometrical isomers.
- For disubstituted benzenes $\mu_{res} = \sqrt{\mu_1^2 + \mu_2^2 + 2\mu_1\mu_2\cos\theta}$ with $\theta = 60°$ (ortho), 120° (meta), 180° (para). If the two substituents are identical, the para isomer has $\mu = 0$ and order is $\mu_{ortho} > \mu_{meta} > \mu_{para}$.

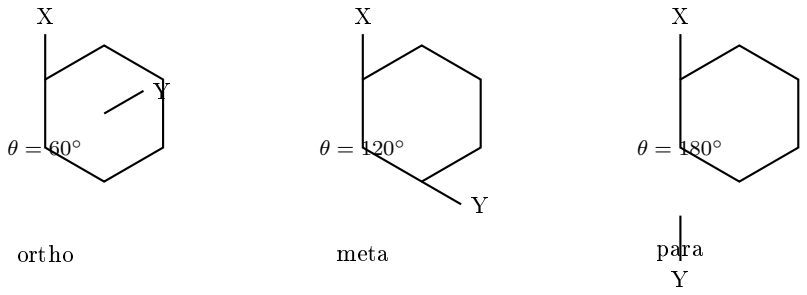


Figure 22. Relative positions of substituents X and Y; the angle θ between the group moments governs the resultant.

17.2 Dipole Moment and Percentage Ionic Character

Formula 17.3 Percentage Ionic Character

$$\%\ \text{ionic character} = \frac{\mu_{\text{observed}}}{\mu_{\text{theoretical}}} \times 100\ , \qquad \mu_{\text{theoretical}} = e \times d$$

Conditions

$\mu_{\text{theoretical}}$ is the moment if the bond were 100% ionic with a full electronic charge ($e = 4.8 \times 10^{-10}$ esu) at the bond length d. Use consistent units: $e\,(4.8 \times 10^{-10}\ \text{esu}) \times d\,(\text{cm}) \div 10^{-18} = \mu$ in D. Applicable to diatomic (or localised single-bond) cases.

Worked Example 17.2 Ionic character of HCl and HF

HCl: $\mu_{\text{obs}} = 1.07$ D, $d = 127$ pm. HF: $\mu_{\text{obs}} = 1.78$ D, $d = 92$ pm.

Solution.

$$\mu_{\text{th}}(\text{HCl}) = 4.8 \times 1.27 = 6.10\ \text{D} \ \Rightarrow\ \% = \frac{1.07}{6.10} \times 100 = 17.5\%$$

$$\mu_{\text{th}}(\text{HF}) = 4.8 \times 0.92 = 4.42\ \text{D} \ \Rightarrow\ \% = \frac{1.78}{4.42} \times 100 = 40.3\%$$

Type	Example	μ (D)	Geometry
AB	HF	1.78	linear
	HCl	1.07	linear
	HBr	0.79	linear
	HI	0.38	linear
A_2	H_2	0	linear
AB_2	H_2O	1.85	bent
	H_2S	0.95	bent
	CO_2	0	linear
AB_3	NH_3	1.47	trigonal pyramidal
	NF_3	0.23	trigonal pyramidal
	BF_3	0	trigonal planar
AB_4	CH_4	0	tetrahedral
	$CHCl_3$	1.04	tetrahedral
	CCl_4	0	tetrahedral

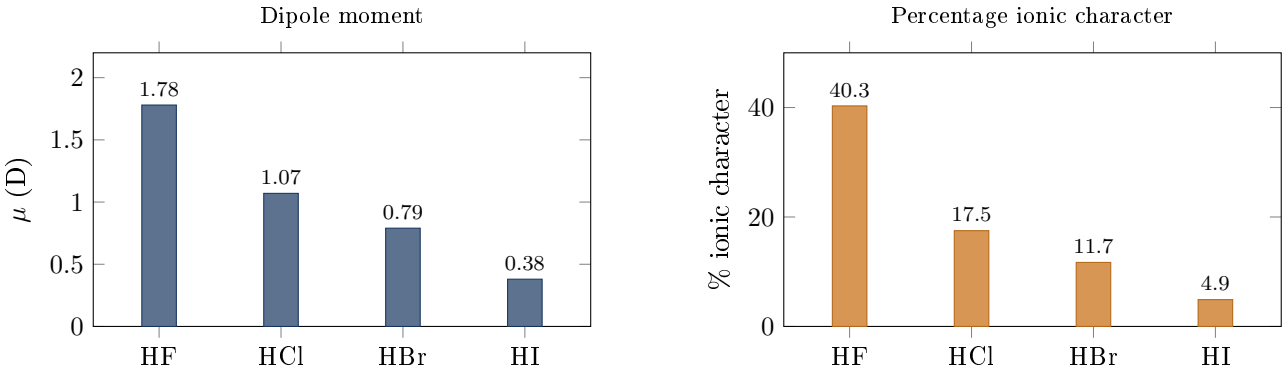


Figure 23. Dipole moments and ionic character of the hydrogen halides (d = 92, 127, 141, 160 pm). Percentages are computed from $\mu_{\text{obs}}/(e\,d)$.

Worked Example 17.3 Correct order of dipole moment

(A) $OF_2 > OH_2$ (B) $NH_3 > SO_2$ (C) $PCl_3F_2 > PCl_2F_3$ (D) $CH_3F < CH_3Cl$
Solution. (A) is wrong: in OF_2 the O–F bond moments oppose the lone-pair moment, whereas in H_2O the O–H moments reinforce it; so $H_2O > OF_2$. (B) is wrong: NH_3 1.47 D $< SO_2$ 1.60 D. (C) is wrong: PCl_3F_2 (F axial, symmetric) is non-polar, PCl_2F_3 is polar. (D) is correct by the notes' data because the very small F makes $q \times d$ smaller than for Cl. Answer **(D)**.

Part VI — Molecular Orbital Theory

18 Molecular Orbital Theory (MOT)

Developed by F. Hund and R. S. Mulliken (1932). Its salient features are:

(i) Electrons of a molecule occupy *molecular orbitals* (MOs), just as atomic electrons occupy atomic orbitals.

(ii) MOs are formed by combining atomic orbitals of comparable energy and proper symmetry.

(iii) An atomic orbital is monocentric (influenced by one nucleus); an MO is polycentric.

(iv) The number of MOs formed equals the number of combining atomic orbitals: two AOs give one *bonding* and one *antibonding* MO.

(v) The bonding MO has lower energy (greater stability) than the antibonding MO.

(vi) The MO describes the electron probability distribution around a group of nuclei.

(vii) MOs are filled according to the Aufbau principle, Pauli's exclusion principle and Hund's rule; the filling order is decided experimentally (there is no $n + l$ rule for MOs).

VBT versus MOT

In VBT the atomic orbitals of one atom are first modified (hybridisation) and then overlapped; in MOT the orbitals of *different atoms* are combined to give a new kind of orbital, delocalised over the molecule.

18.1 Formation of Molecular Orbitals: Linear Combination of Atomic Orbitals (LCAO)

The Schrödinger equation cannot be solved exactly for a molecule of more than one electron, so molecular wave functions are approximated by a linear combination of atomic orbitals. For H_2 (atoms A and B, wave functions ψ_A, ψ_B):

Formula 18.1 LCAO Wave Functions

$$\psi_\sigma = \psi_\mathrm{A} + \psi_\mathrm{B} \quad \text{(bonding)}\ , \qquad \psi_{\sigma^*} = \psi_\mathrm{A} - \psi_\mathrm{B} \quad \text{(antibonding)}$$

Derivation

Electron density. Squaring the wave functions gives the probability density:

$$\psi_\sigma^2 = \psi_\mathrm{A}^2 + \psi_\mathrm{B}^2 + 2\psi_\mathrm{A}\psi_\mathrm{B}\ , \qquad \psi_{\sigma^*}^2 = \psi_\mathrm{A}^2 + \psi_\mathrm{B}^2 - 2\psi_\mathrm{A}\psi_\mathrm{B}$$

The cross-term $2\psi_\mathrm{A}\psi_\mathrm{B}$ *adds* density between the nuclei in σ (constructive interference, nuclear repulsion screened, energy lowered) and *removes* it in σ^* (destructive interference, a nodal plane between nuclei).
Energies. With α the Coulomb integral (energy of an electron in an isolated AO, $\alpha < 0$), β the resonance integral ($\beta < 0$) and S the overlap integral,

$$E_\sigma = \frac{\alpha + \beta}{1 + S}\ , \qquad E_{\sigma^*} = \frac{\alpha - \beta}{1 - S}$$

Since $1 - S < 1 + S$, the antibonding level is raised above α by *more* than the bonding level is lowered below it, but the total for the two MOs equals that of the two original AOs when $S \to 0$.

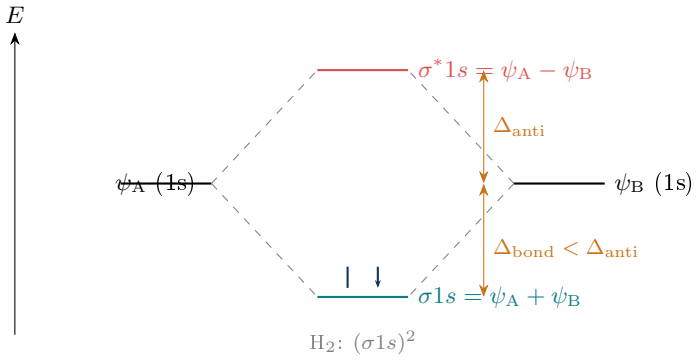


Figure 24. Formation of bonding (σ) and antibonding (σ^*) MOs by LCAO of two 1s orbitals. In practice Δ_anti slightly exceeds Δ_bond.

18.2 Conditions for the Combination of Atomic Orbitals

(i) **Comparable energy.** 1s can combine with 1s but not with 2s.

(ii) **Same symmetry about the molecular axis** (by convention the z axis): $2\mathrm{p}_z$ combines with $2\mathrm{p}_z$, never with $2\mathrm{p}_x$ or $2\mathrm{p}_y$.

(iii) **Maximum overlap.** The greater the overlap, the greater the electron density between the nuclei and the larger the energy lowering.

18.3 Types of Molecular Orbitals

- σ **MOs** are symmetrical about the bond axis: from s–s ($\sigma 1s, \sigma^* 1s$), p_z–p_z end-on ($\sigma 2p_z, \sigma^* 2p_z$) and s–$\mathrm{p}_z$ combinations ($\sigma 2s, \sigma^* 2s$ in the presence of s–p mixing).
- π **MOs** arise from p_x–p_x and p_y–p_y sideways overlap; they are not symmetrical about the axis and have a nodal plane containing the axis ($\pi 2p_x, \pi 2p_y, \pi^* 2p_x, \pi^* 2p_y$).
- δ **MOs** involve d orbitals (not required for second-period diatomics).

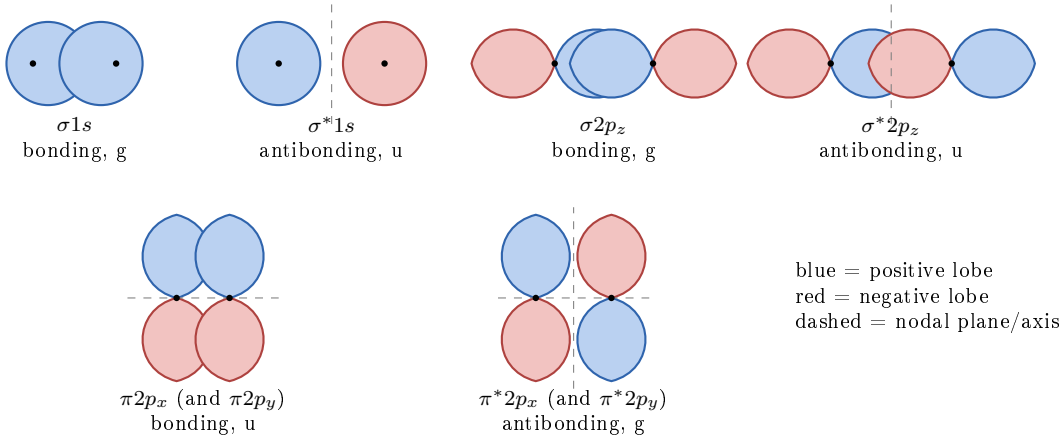


Figure 25. Shapes of the principal molecular orbitals of a homonuclear diatomic. The labels g (gerade) and u (ungerade) refer to symmetry on inversion through the molecular centre.

MO	Origin	Nodal plane between nuclei?	Symmetry
$\sigma 1s$, $\sigma 2s$, $\sigma 2p_z$	in-phase	no	g
$\sigma^* 1s$, $\sigma^* 2s$, $\sigma^* 2p_z$	out-of-phase	yes (perpendicular to axis)	u
$\pi 2p_x$, $\pi 2p_y$	in-phase	one (contains axis)	u
$\pi^* 2p_x$, $\pi^* 2p_y$	out-of-phase	two	g

g and u test. Invert the orbital through the centre of the molecule: if the sign of each lobe is unchanged the MO is gerade (g, even); if the signs change it is ungerade (u, odd).

18.4 Energy Level Diagram for Molecular Orbitals

The two sets of 2s and 2p AOs (eight orbitals) give eight MOs: $\sigma 2s, \sigma^* 2s, \sigma 2p_z, \pi 2p_x, \pi 2p_y, \pi^* 2p_x, \pi^* 2p_y, \sigma^* 2p_z$. The order has been fixed from spectroscopic data.

$$\boxed{\mathrm{O_2,\ F_2,\ Ne_2}: \quad \sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z}$$

$$\boxed{\mathrm{Li_2,\ Be_2,\ B_2,\ C_2,\ N_2}: \quad \sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_y) < \sigma 2p_z < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z}$$

Why the order changes: s–p mixing

In the lighter molecules (up to N_2) the 2s–2p energy gap is small, so the $\sigma 2s/\sigma 2p_z$ (and $\sigma^* 2s/\sigma^* 2p_z$) MOs of the same symmetry mix. This pushes $\sigma 2p_z$ *up* above the $\pi 2p$ pair. In O_2 and F_2 the 2s–2p gap is large, mixing is negligible, and the "normal" order ($\sigma 2p_z$ below $\pi 2p$) is observed.

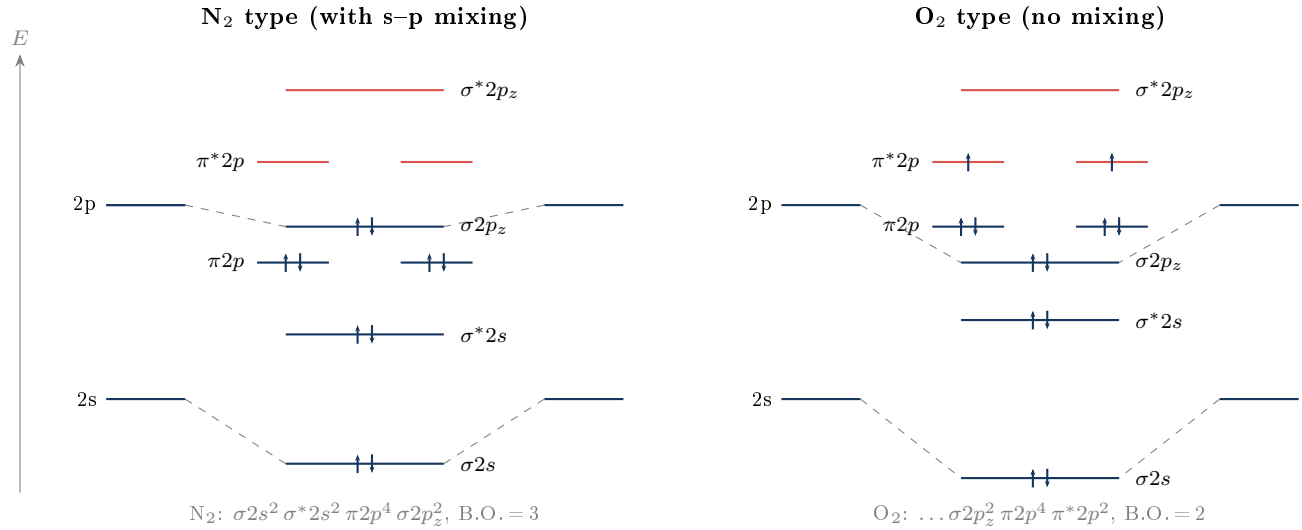


Figure 26. Valence molecular-orbital energy diagrams (the 1s core MOs are omitted). Left: the $\pi 2p$ pair lies below $\sigma 2p_z$ because of s–p mixing (Li_2 to N_2). Right: for O_2 and F_2 the order is reversed; the two electrons in π^*2p occupy separate orbitals with parallel spins, making O_2 paramagnetic.

18.5 Electronic Configuration and Molecular Behaviour

Formula 18.2 Bond Order

$$\boxed{\text{Bond order (B.O.)} = \frac{N_b - N_a}{2}}$$

Conditions

N_b = number of electrons in bonding MOs, N_a = number in antibonding MOs (count all electrons, or valence electrons only: the core $\sigma 1s, \sigma^* 1s$ cancel). If $N_b > N_a$ (B.O. positive) the molecule is stable; if $N_b \le N_a$ (zero or negative B.O.) it is unstable.

Formula 18.3 Properties from Bond Order

$$\boxed{\text{B.O. } \uparrow \Longrightarrow \text{ bond energy } \uparrow, \quad \text{bond length } \downarrow, \quad \text{stability } \uparrow}$$

$$\boxed{\mu_{\text{spin}} = \sqrt{n(n+2)}\ \text{B.M.}}$$

Conditions

Integral B.O. $= 1, 2, 3$ corresponds to single, double, triple bond. If every MO is doubly occupied ($n = 0$) the substance is **diamagnetic**; if one or more MOs are singly occupied (n unpaired electrons) it is **paramagnetic**. Spin-only formula; n = number of unpaired electrons.

18.6 Bonding in Some Homonuclear Diatomic Molecules

Species	Configuration (valence MOs follow KK or $\sigma 1s^2\sigma^*1s^2$)	B.O.	Unpaired e^-	Magnetic
$\mathbf{H_2}$	$(\sigma 1s)^2$	1	0	diamagnetic
$\mathbf{He_2}$	$(\sigma 1s)^2(\sigma^*1s)^2$	0	0	does not exist
$\mathbf{Li_2}$	$\mathrm{KK}(\sigma 2s)^2$	1	0	diamagnetic
$\mathbf{Be_2}$	$\mathrm{KK}(\sigma 2s)^2(\sigma^*2s)^2$	0	0	does not exist
$\mathbf{B_2}$	$\mathrm{KK}(\sigma 2s)^2(\sigma^*2s)^2(\pi 2p_x^1 = \pi 2p_y^1)$	1	2	paramagnetic
$\mathbf{C_2}$	$\mathrm{KK}(\sigma 2s)^2(\sigma^*2s)^2(\pi 2p_x^2 = \pi 2p_y^2)$	2	0	diamagnetic
$\mathbf{N_2}$	$\mathrm{KK}(\sigma 2s)^2(\sigma^*2s)^2(\pi 2p_x^2 = \pi 2p_y^2)(\sigma 2p_z)^2$	3	0	diamagnetic
$\mathbf{O_2}$	$\mathrm{KK}(\sigma 2s)^2(\sigma^*2s)^2(\sigma 2p_z)^2(\pi 2p_x^2 = \pi 2p_y^2)(\pi^*2p_x^1 = \pi^*2p_y^1)$	2	2	paramagnetic
$\mathbf{F_2}$	$\ldots(\sigma 2p_z)^2(\pi 2p)^4(\pi^*2p)^4$	1	0	diamagnetic
$\mathbf{Ne_2}$	$\ldots(\pi^*2p)^4(\sigma^*2p_z)^2$	0	0	does not exist

Note

In C_2 the double bond consists of *two π bonds* (four electrons in the two $\pi 2p$ MOs) and no σ bond other than that from $\sigma 2s/\sigma^*2s$ cancellation; in most other molecules a double bond is one σ plus one π bond. The $\sigma 1s$ and σ^*1s pair (KK) is the closed K-shell.

Property	B_2	C_2	N_2	O_2	F_2
Bond order	1	2	3	2	1
Bond energy (kJ mol^{-1})	290	620	945	498	159
Bond length (pm)	159	131	110	121	143
Magnetism	para	dia	dia	para	dia

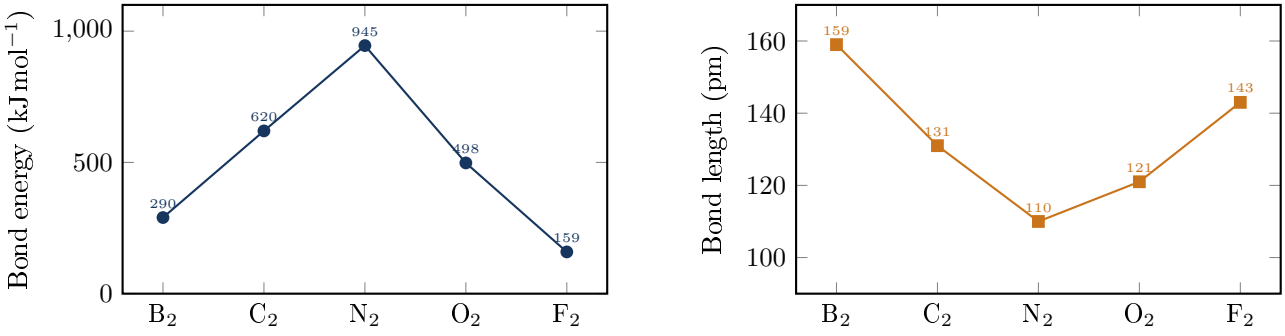


Figure 27. Bond energy and bond length across B_2 to F_2: the maximum energy and minimum length coincide with the maximum bond order (N_2, B.O. = 3).

Ions derived from N_2 and O_2

- $\mathbf{N_2^-}$ (15 e$^-$) has the configuration of N_2 plus one electron in π^*2p_x: B.O. $= \frac{1}{2}(10-5) = 2.5$. N_2^+ also has B.O. 2.5, but N_2^- is the *less stable* of the two because it has more antibonding electrons.
- O_2^+: B.O. 2.5 (paramagnetic); O_2^-: 1.5 (paramagnetic); O_2^{2-}: 1 (diamagnetic).

	O_2^{2+}	O_2^+	O_2	O_2^-	O_2^{2-}
Valence electrons ($\sigma/\pi/\pi^*$ filling)	10	11	12	13	14
Unpaired electrons	0	1	2	1	0
Bond order	3.0	2.5	2.0	1.5	1.0
Bond length (pm)	–	112	121	128	149
Magnetism	dia	para	para	para	dia

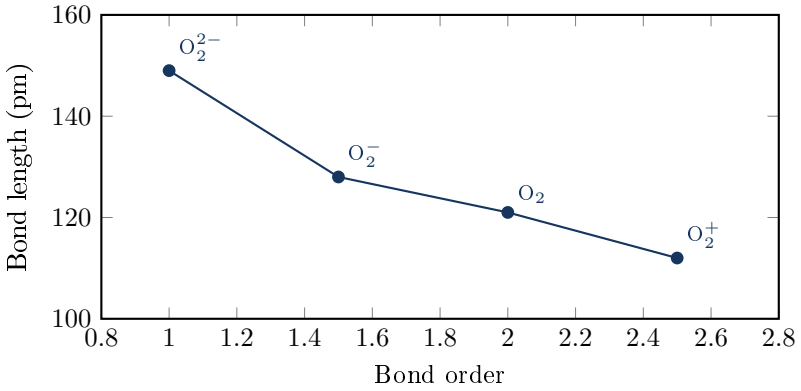


Figure 28. Bond length falls as bond order rises for the O_2 family.

Worked Example 18.1 Order of bond lengths: O_2, O_2^+, O_2^-, O_2^{2-}

Solution. Bond length $\propto 1/\text{B.O.}$; B.O.: $O_2^+ = 2.5$, $O_2 = 2$, $O_2^- = 1.5$, $O_2^{2-} = 1$. Therefore

$$\boxed{O_2^+ < O_2 < O_2^- < O_2^{2-}}$$

Worked Example 18.2 Antibonding electron pairs in O_2^{2-}

Solution. O_2^{2-} has $2(8) + 2 = 18$ electrons: $\sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p^4 \pi^* 2p^4$. Antibonding electrons $N_a = 2 + 2 + 4 = 8$, i.e. **4 pairs** (answer C). B.O. $= \frac{1}{2}(10 - 8) = 1$.

Worked Example 18.3 Increasing bond order

Arrange NO, C_2, O_2^-, He_2^+.
Solution. B.O.: $He_2^+ = \frac{1}{2}(2 - 1) = 0.5$; $O_2^- = 1.5$; $C_2 = 2$; NO (15 e$^-$, N_2-type order) $= \frac{1}{2}(10 - 5) = 2.5$.

$$\boxed{He_2^+ < O_2^- < C_2 < NO}$$

Worked Example 18.4 Species isoelectronic with CO and with the same bond order

CO has 14 electrons and B.O. 3 (isoelectronic with N_2). Among CN^-, O_2^+, O_2^-, N_2^+ only $\mathbf{CN^-}$ (14 e^-) matches. Diamagnetic isoelectronic partners of N_2: CO, CN^-, NO^+, C_2^{2-} (B.O. 3).

Worked Example 18.5 Spin-only magnetic moment of O_2 and B_2

Both have $n = 2$ unpaired electrons:

$$\boxed{\mu = \sqrt{2(2+2)} = \sqrt{8} = 2.83 \text{ B.M.}}$$

JEE Pointer

Quick MO checklist: (1) count total electrons; (2) choose the order (N_2-type for total $e^- \leq 14$ in B_2–N_2 series and their ions; O_2-type for O_2, F_2 and ions); (3) fill with Hund's rule in degenerate sets; (4) B.O. $= \frac{1}{2}(N_b - N_a)$; (5) unpaired electrons give magnetism. Paramagnetic: B_2, O_2, $O_2^{\pm}$, NO, $N_2^{\pm}$; Diamagnetic: C_2, N_2, F_2, O_2^{2-}, CO, CN^-.

Part VII — Exam Tools

19 Master Formula Atlas

Family	Formula	Condition / keyword
Born–Haber	$\Delta H_f = S + \mathrm{IE} + \frac{1}{2}D - \mathrm{EGE} - U$	ionic solid; EGE, U as energies released
Born–Haber (MX_2)	$\Delta H_f = S + \mathrm{IE}_1 + \mathrm{IE}_2 + D - 2\,\mathrm{EGE} - U$	divalent metal, halogen
Lattice energy	$U \propto z^+ z^- / (r^+ + r^-)$	higher charge, smaller ions
Solubility	$\Delta H_{\mathrm{sol}} = U - \lvert\Delta H_{\mathrm{hyd}}\rvert$	dissolves if hydration > lattice
Ionic potential	$\phi = z^+ / r^+$	polarising power; Fajans' rule
Covalent character	$\propto \phi_{\mathrm{cation}} \times \mathrm{polarisability}_{\mathrm{anion}}$	$18e^-$ cation > $8e^-$ cation
Valence electrons	$N_{\mathrm{val}} = \sum V_i - q$	Lewis structures
Formal charge	$\mathrm{F.C.} = V - L - \frac{1}{2}B$	lowest $\sum \lvert\mathrm{F.C.}\rvert$ preferred
Hybrid-orbital count	$H = \frac{1}{2}[V + M - C + A]$	$2, 3, 4, 5, 6, 7 \rightarrow$ sp, sp^2, sp^3, sp^3d, sp^3d^2, sp^3d^3
Method I	$H = \#\sigma + \#\mathrm{l.p.}$	central atom
s-Character	$\%s = 100/H$	equivalent hybrids
σ/π count	single = 1σ; double = $1\sigma + 1\pi$; triple = $1\sigma + 2\pi$	orbital overlap
Bond length	$d_{\mathrm{A-B}} = r_{\mathrm{A}} + r_{\mathrm{B}} - 0.09\lvert\Delta\chi\rvert$	Å; polar single bond
Dipole moment	$\mu = q \times d$; $1\,\mathrm{D} = 3.336 \times 10^{-30}$ C m	polarity
Resultant moment	$\mu = \sqrt{\mu_1^2 + \mu_2^2 + 2\mu_1\mu_2\cos\theta}$	θ = angle between vectors
Ionic character	$\% = \mu_{\mathrm{obs}}/(e\,d) \times 100$	$e = 4.8 \times 10^{-10}$ esu
LCAO	$\psi_\sigma = \psi_A + \psi_B$; $\psi_{\sigma^*} = \psi_A - \psi_B$	H_2-type
Bond order	$\mathrm{B.O.} = (N_b - N_a)/2$	MOT
Magnetic moment	$\mu = \sqrt{n(n+2)}$ B.M.	n unpaired electrons

20 Keyword Decision Tree

- "**atoms combine**", "**energy lowered**" → bond formation is exothermic; attraction > repulsion.
- "**transfer of electrons**", "**non-directional**", "**no molecular formula**" → ionic.
- "**covalent character**", "**colour**", "**thermal stability of carbonate**" → Fajans/polarising power: z^+/r^+ and $18e^-$ cations.
- "**solubility in water**" → compare lattice and hydration energy; "**non-polar solvent**" → covalent compound.
- "**shape of molecule**" → count b.p. + l.p.; lone pairs go equatorial in TBP.
- "**hybridisation of central atom**" → $\#\sigma + \#$l.p., or $H = \frac{1}{2}[V + M - C + A]$.
- "**bond length**" → size, s-character, bond order, resonance; "**bond angle**" → hybridisation, lone pairs, χ of central atom, size of terminal atoms.
- "**bond angle near 90° in PH_3, H_2S**" → Drago generalisation.
- "**electronegative substituent position**", "**lone pair in hybrid**" → Bent's rule.
- "**dipole moment**" → symmetry first (vector sum), then $\mu = q \times d$.
- "**paramagnetic**", "**bond order**", "**isoelectronic**" → MOT; check N_2-type vs O_2-type order.
- "**species does not exist**" → no d-orbitals (period 2), steric crowding, or less electronegative ligand (H).

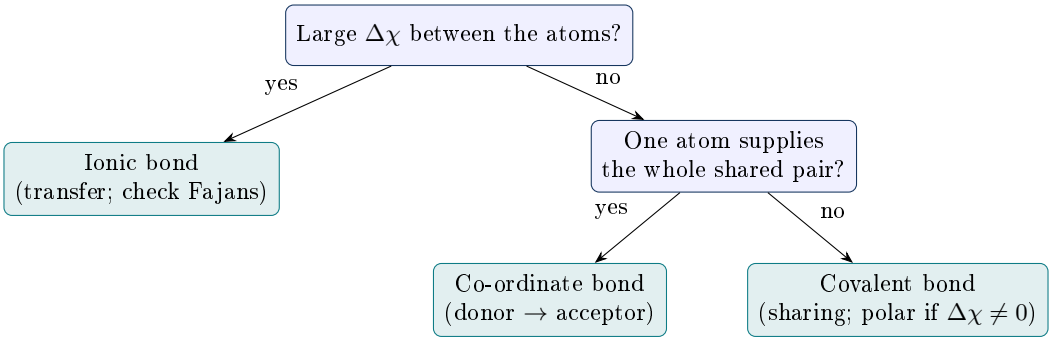


Figure 29. Quick decision flow for the type of bond (strong-bond classes).

21 Summary of Trends in Bonding

Property	Order
Lattice energy	$LiCl > NaCl > KCl > RbCl > CsCl$; $NaCl < MgCl_2 < AlCl_3$ (ionic model)
Covalent character	$BeCl_2 > MgCl_2 > CaCl_2 > SrCl_2$; $AgCl < AgBr < AgI$; $MnO < MnO_2 < Mn_2O_7$
Thermal stability (carbonates)	$BeCO_3 < MgCO_3 < CaCO_3 < SrCO_3 < BaCO_3$
C–C bond length	sp^3–$sp^3 > sp^2$–$sp^2 > sp$–sp
Bond energy	C–C < C=C < C≡C; HF > HCl > HBr > HI; Cl–Cl > Br–Br > I–I
Bond angle	$BeH_2 > BH_3 > CH_4 > NH_3 > H_2O$; $H_2O > H_2S > H_2Se > H_2Te$
Dipole moment	$CH_3Cl > CH_2Cl_2 > CHCl_3 > CCl_4 = 0$; HF > HCl > HBr > HI
Bond length (O_2 family)	$O_2^+ < O_2 < O_2^- < O_2^{2-}$

22 Last-Minute Exam Traps

Traps to avoid

- Period-2 central atoms never exceed an octet (NO_3^- has three σ bonds; N is sp^2; NF_5 and OF_6 do not exist).
- Hybridisation counts σ bonds plus lone pairs; π bonds do not count.
- In TBP structures lone pairs are equatorial; more electronegative ligands go axial (Bent's rule).
- Drago's rule applies only to period ≥ 3 central atoms with $\chi_{\text{ligand}} \leq 2.5$; NH_3 and H_2O are hybridised and obey VSEPR.
- A polar bond is not a polar molecule: check symmetry (CO_2, BF_3, CCl_4, XeF_4, SF_6, PCl_5 are non-polar).
- N_2^- and N_2^+ share B.O. 2.5 but N_2^- is less stable.
- For B_2, C_2, N_2 use the π-below-σ order; for O_2, F_2 use σ below π.
- Lattice energy and electron gain enthalpy are entered with opposite signs to ionisation and sublimation terms in Born–Haber work.
- F_2 bond is weaker than Cl_2 despite the smaller size (lone-pair repulsion); bond length of F_2 is 144 pm.
- Solubility trend down Group 2: sulphates fall, hydroxides rise.

23 Symbol and Sign Glossary

Symbol	Meaning
U	lattice energy (energy released on forming one mole of ionic solid from gaseous ions)
S, D, IE, EGE	enthalpy of sublimation; bond dissociation enthalpy; ionisation energy; electron gain enthalpy
ϕ	ionic potential z/r
V, L, B	valence, lone-pair (non-bonding) and bonding electrons in formal charge
H	number of hybrid orbitals
M, C, A	monovalent atoms, cationic charge, anionic charge in $H = \frac{1}{2}[V + M - C + A]$
σ, π, δ	types of covalent bond or molecular orbital
σ^*, π^*	antibonding MOs
N_b, N_a	electrons in bonding, antibonding MOs
μ	dipole moment (D) or magnetic moment (B.M.) by context
χ	Pauling electronegativity

Unit reminders. 1 eV = 96.485 kJ mol^{-1}; 1 kcal = 4.184 kJ; 1 Å = 100 pm; 1 D = 3.336×10^{-30} C m; $e = 4.8 \times 10^{-10}$ esu.

End of the Chemical Bonding atlas. Every boxed result is traceable to a derivation or justification given above.